

Walmart Sales Forecasting and Optimization

Overview

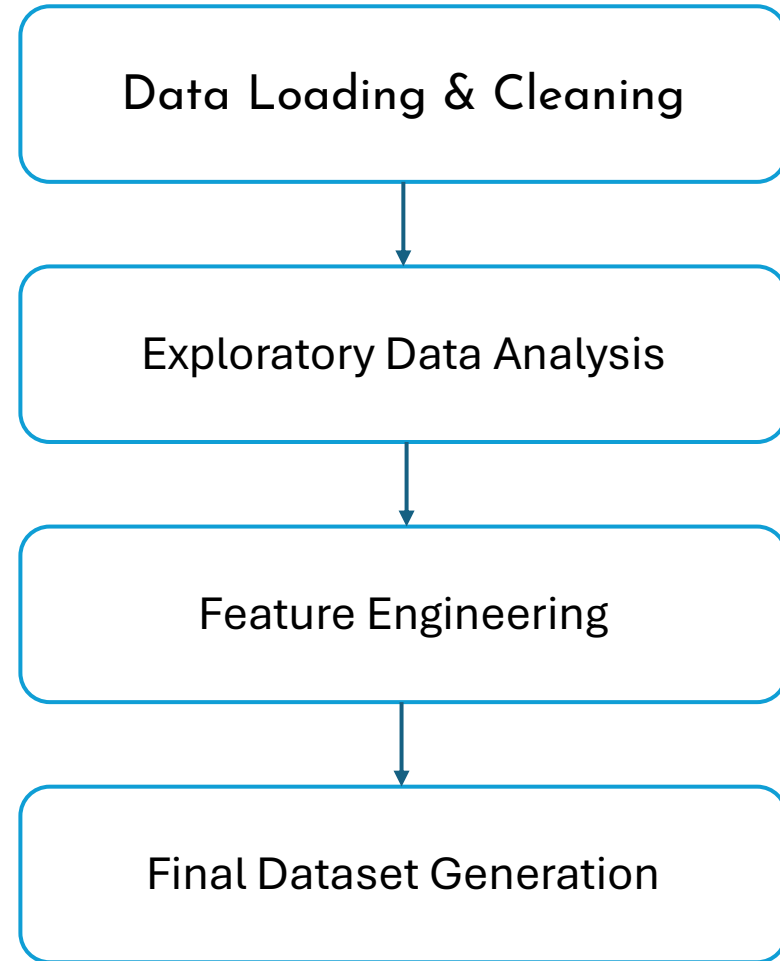
This project develops a machine learning system to forecast weekly sales for Walmart stores across 45 locations and 99 departments. The goal is to predict future sales using historical data, store characteristics, external factors, and promotional activities.

Milestone 1: Data Preprocessing & Feature Engineering

In this milestone, our main goal was to transform raw historical Walmart data into a clean, rich, feature-engineered dataset that is ready for forecasting.

Stage 1 forms the foundation for all later modeling and analysis.

Milestone 1: Overview of Stage 1



Milestone 1: Raw Data Inputs

Our input data included historical weekly sales, store information, and external factors such as CPI, fuel price, and markdown promotions. The challenge was merging all this information into a unified structure.



train.csv
(421,570 rows)



test.csv
(115,064 rows)



features.xlsx

Temperature
Fuel Price
Markdown



stores.csv

Type
Size

Step 1.1: Data Loading & Merging

In Step 1.1, we merged all datasets into consolidated training and test files. This step ensured that every sales record now includes the corresponding store information and external features.

After merging, we had 16 columns in training data and 15 in test data.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Store	Dept	Date	Weekly_Sa	IsHoliday	Type	Size	Temperatu	Fuel_Price	MarkDown	MarkDown	MarkDown	MarkDown	MarkDown	CPI	Unemployment	
2	1	1	5/02/2010	24924.5	FALSE	A	151315	42.31	2.572						211.0964	8.106	
3	1	1	12/02/2010	46039.49	TRUE	A	151315	38.51	2.548						211.2422	8.106	
4	1	1	19/02/2010	41595.55	FALSE	A	151315	39.93	2.514						211.2891	8.106	
5	1	1	26/02/2010	19403.54	FALSE	A	151315	46.63	2.561						211.3196	8.106	
6	1	1	5/03/2010	21827.9	FALSE	A	151315	46.5	2.625						211.3501	8.106	
7	1	1	12/03/2010	21043.39	FALSE	A	151315	57.79	2.667						211.3806	8.106	
8	1	1	19/03/2010	22136.64	FALSE	A	151315	54.58	2.72						211.2156	8.106	
9	1	1	26/03/2010	26229.21	FALSE	A	151315	51.45	2.732						211.018	8.106	
10	1	1	2/04/2010	57258.43	FALSE	A	151315	62.27	2.719						210.8204	7.808	
11	1	1	9/04/2010	42960.91	FALSE	A	151315	65.86	2.77						210.6229	7.808	
12	1	1	16/04/2010	17596.96	FALSE	A	151315	66.32	2.808						210.4887	7.808	
13	1	1	23/04/2010	16145.35	FALSE	A	151315	64.84	2.795						210.4391	7.808	

Step 1.2: Handling Missing Values

Markdown columns had high missing percentages—up to 74%.

Instead of dropping them, we filled missing values with 0 and created binary indicator features to signal whether a markdown existed.

This not only kept the data but also enriched it.

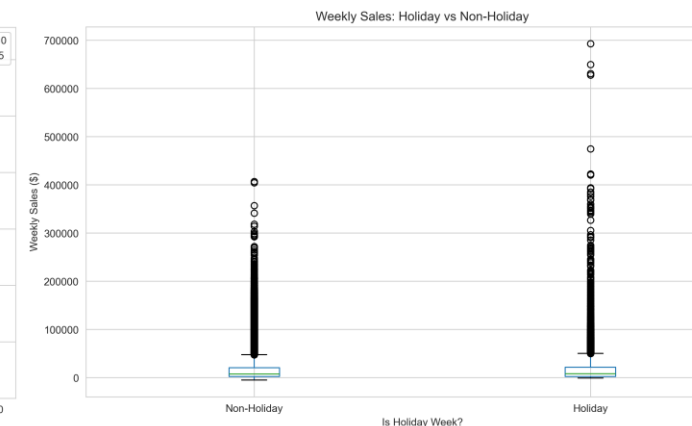
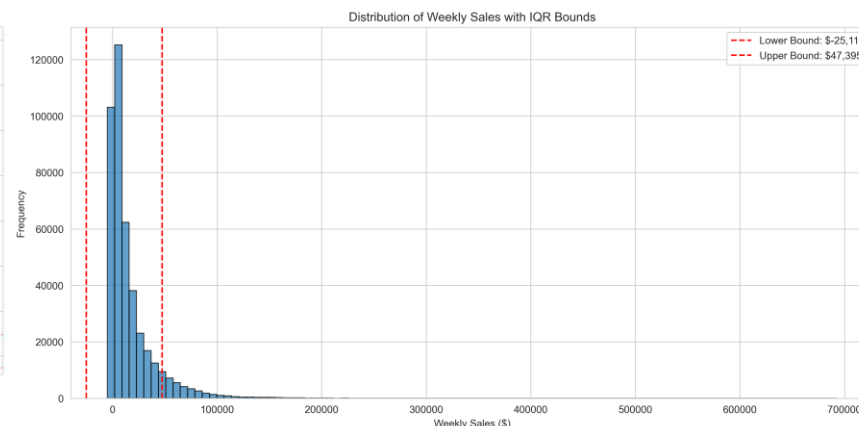
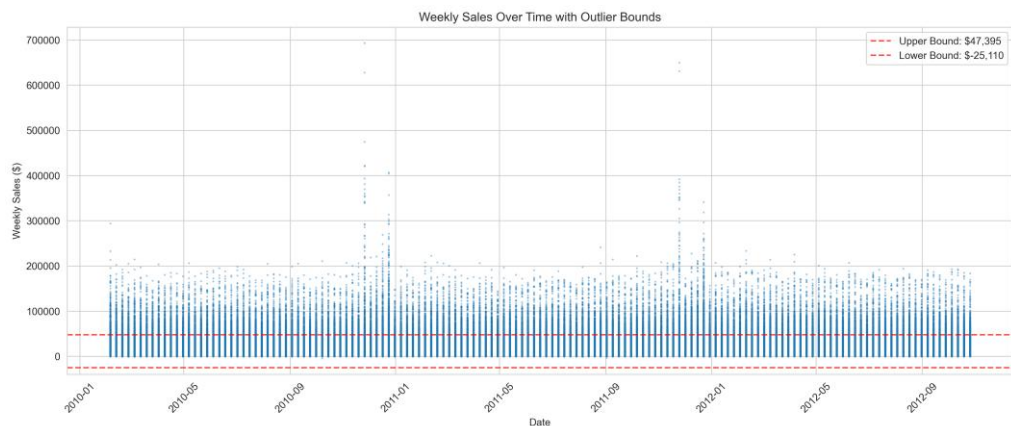
By the end of this step, we achieved 100% completeness with **zero missing values**.

[illegible][illegible]

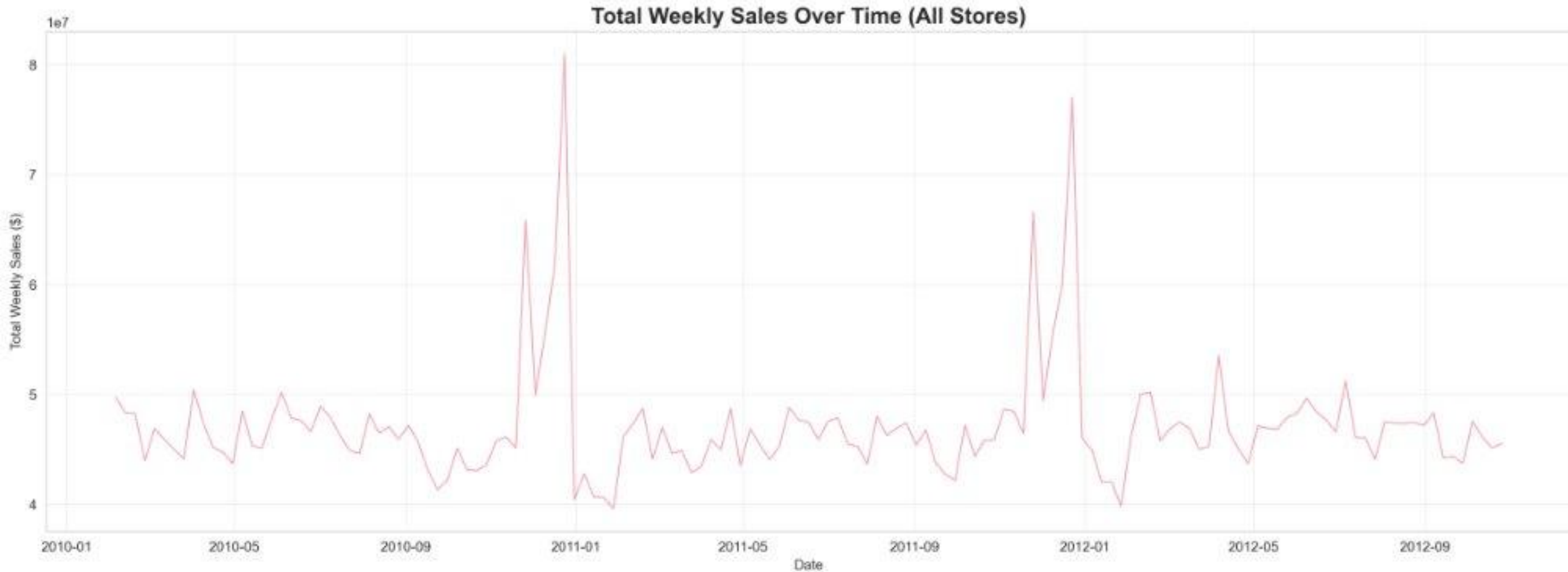
Step 1.3: Outlier Detection

We analyzed outliers using the IQR method and plotted distributions of sales. Although outliers existed, they represented real business events such as holiday spikes. We decided to keep all outliers to avoid removing meaningful patterns.

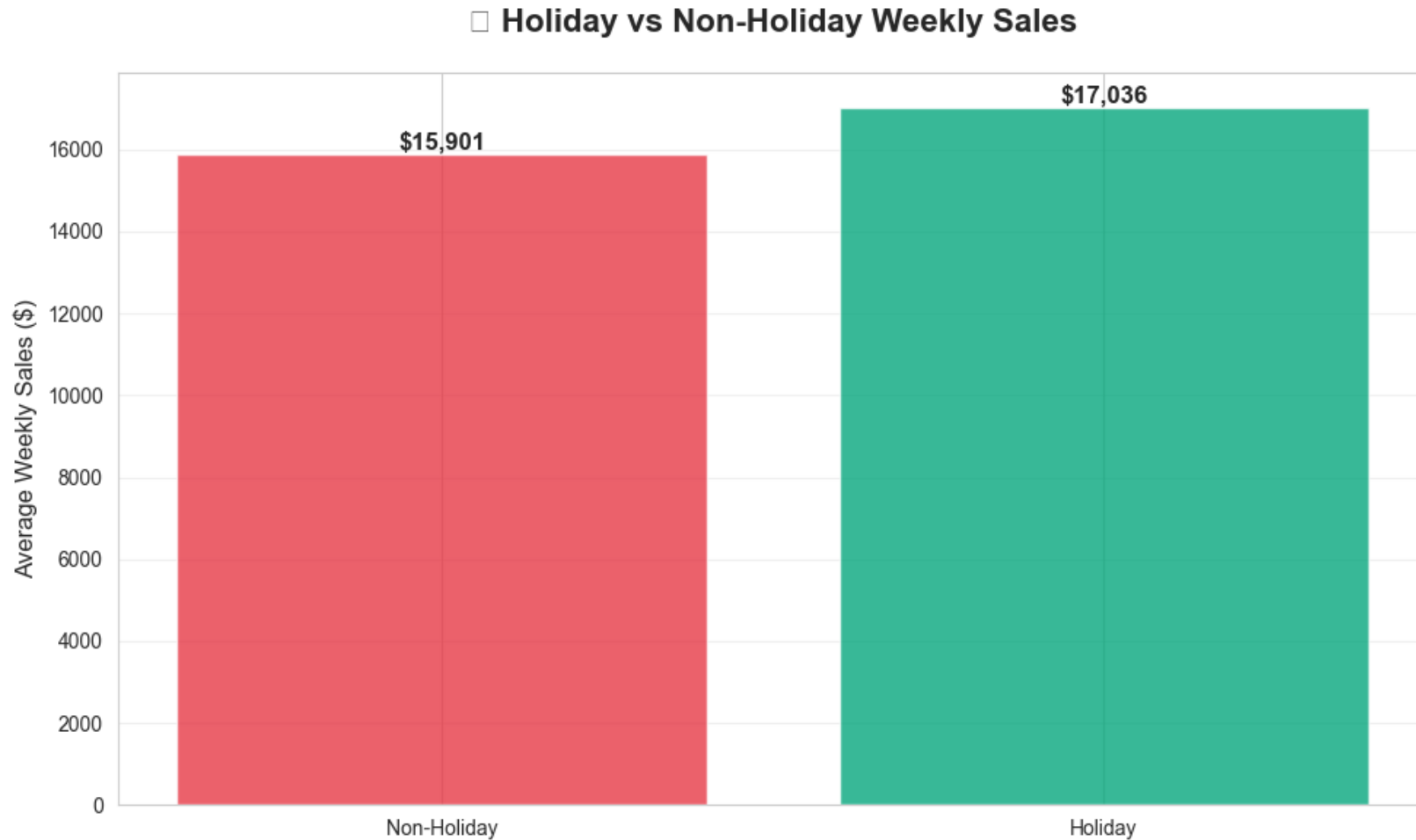
IQR method: $Q3 - Q1$ (middle half of the data).



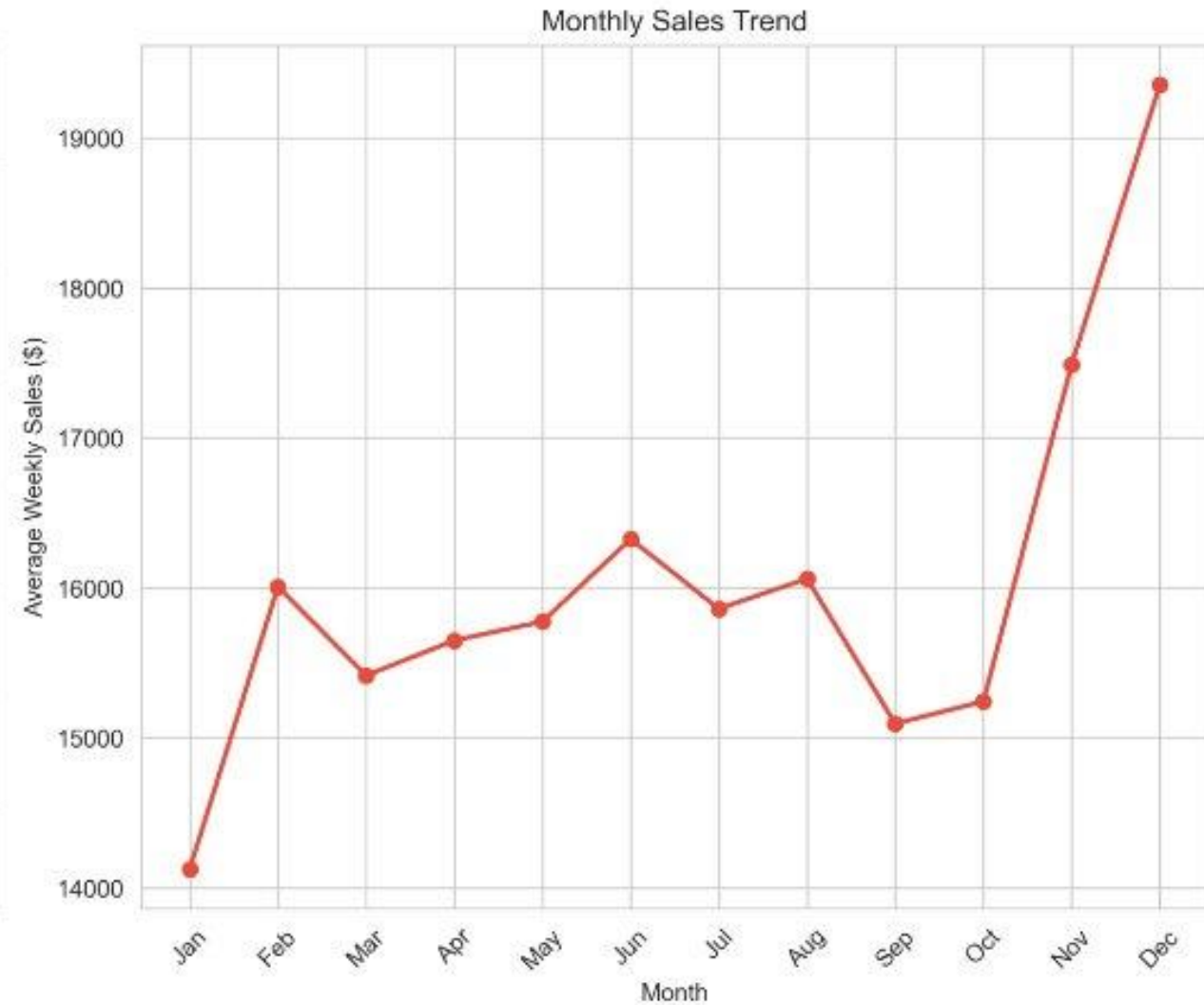
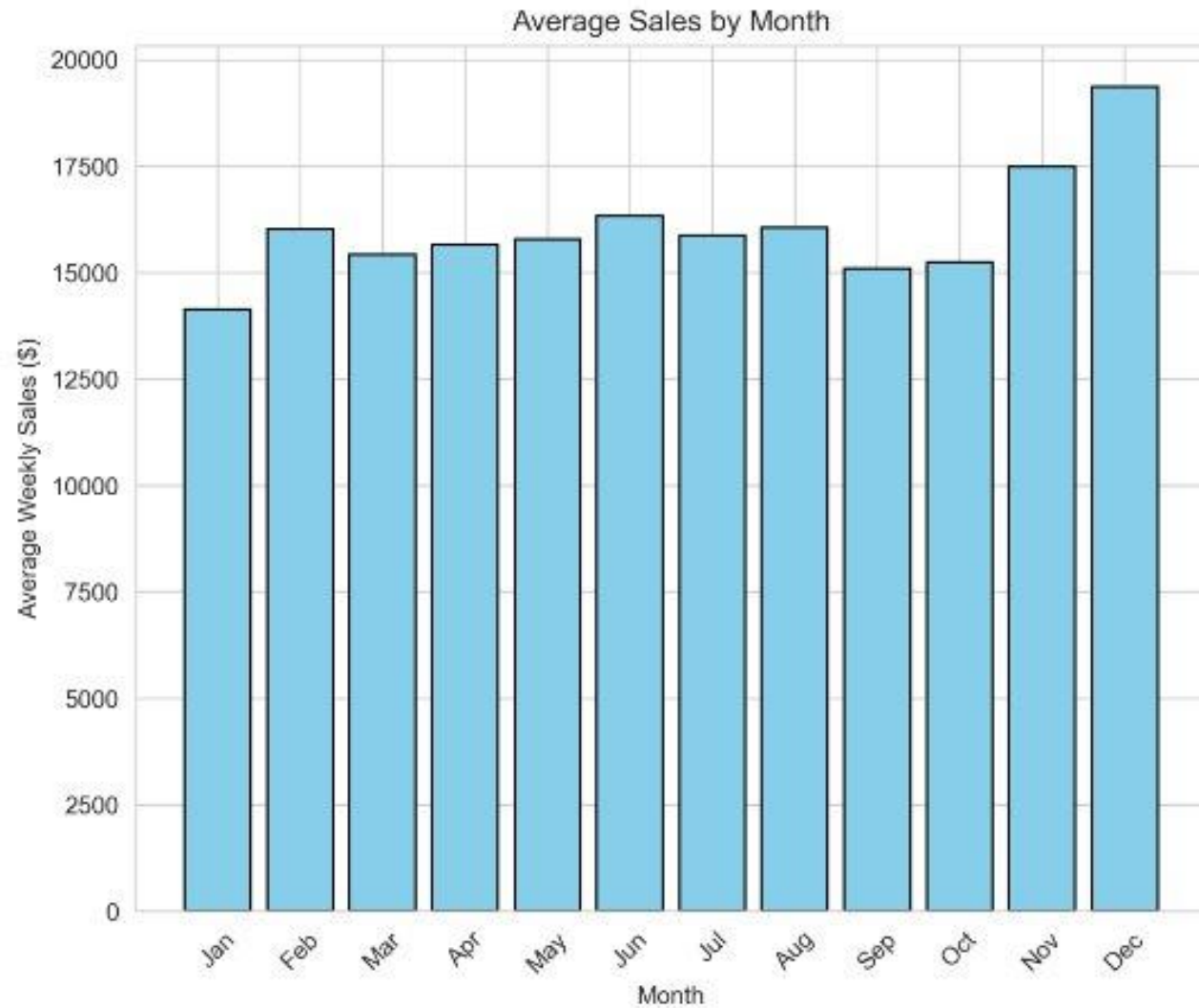
Step 1.4: EDA – Understanding the Data



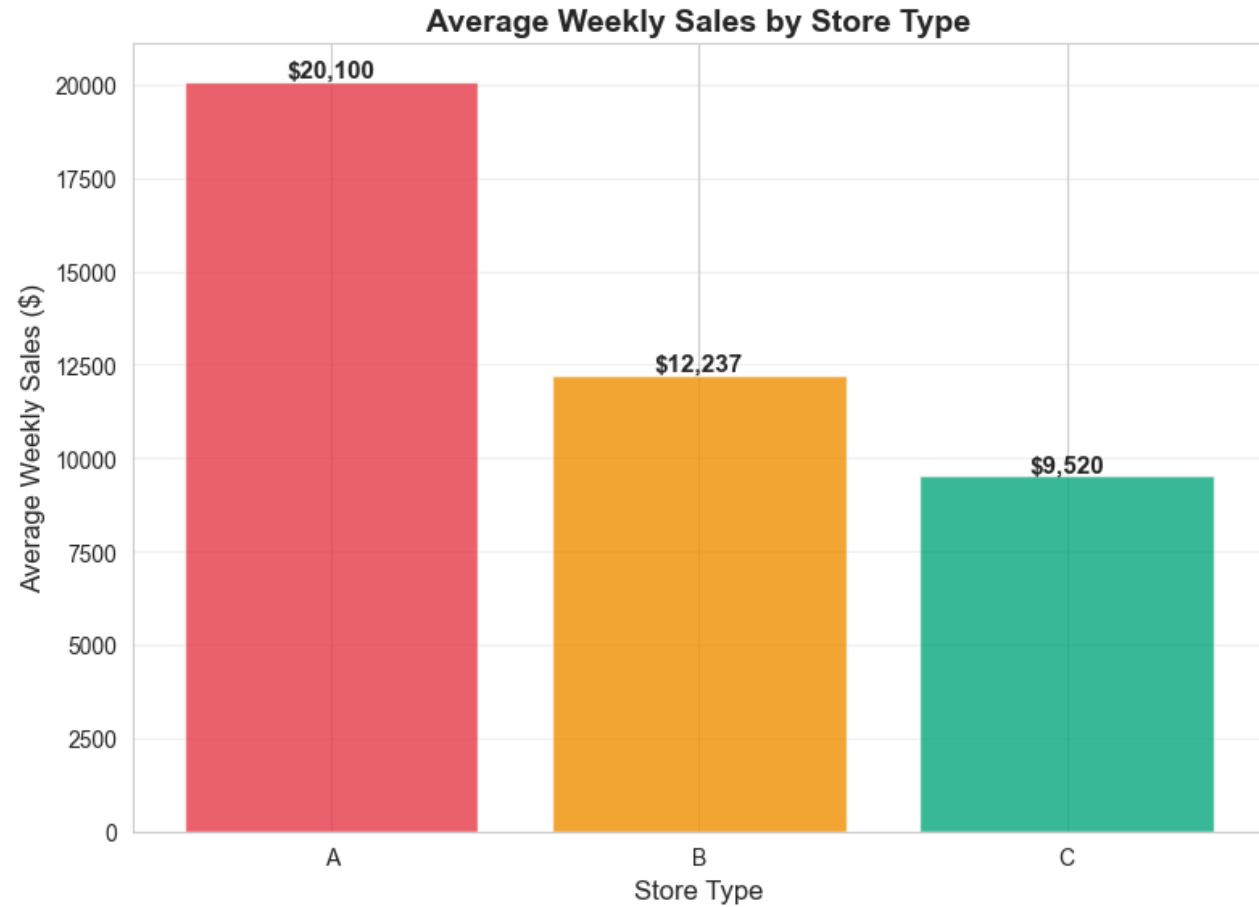
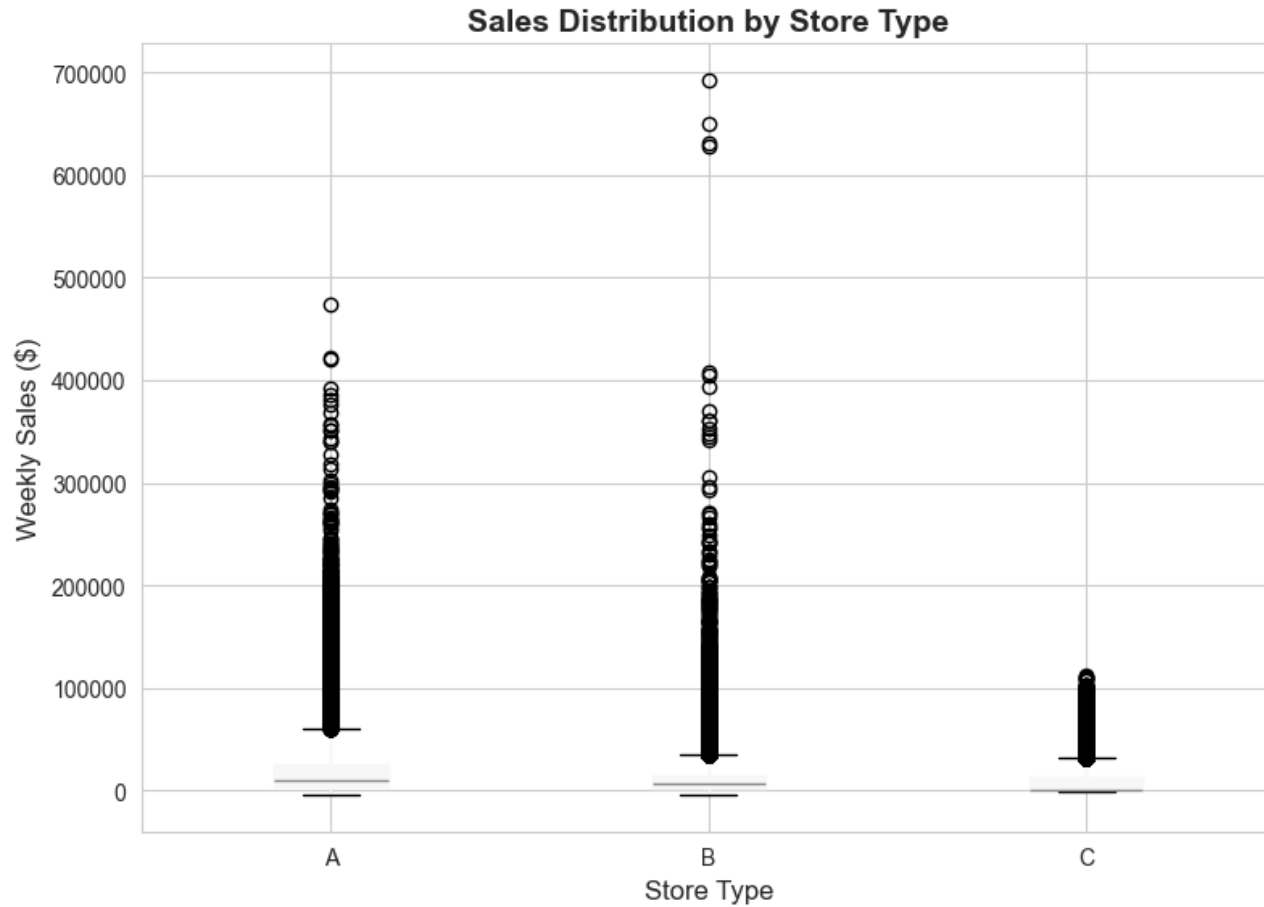
Step 1.4: EDA – Understanding the Data



Step 1.4: EDA – Understanding the Data

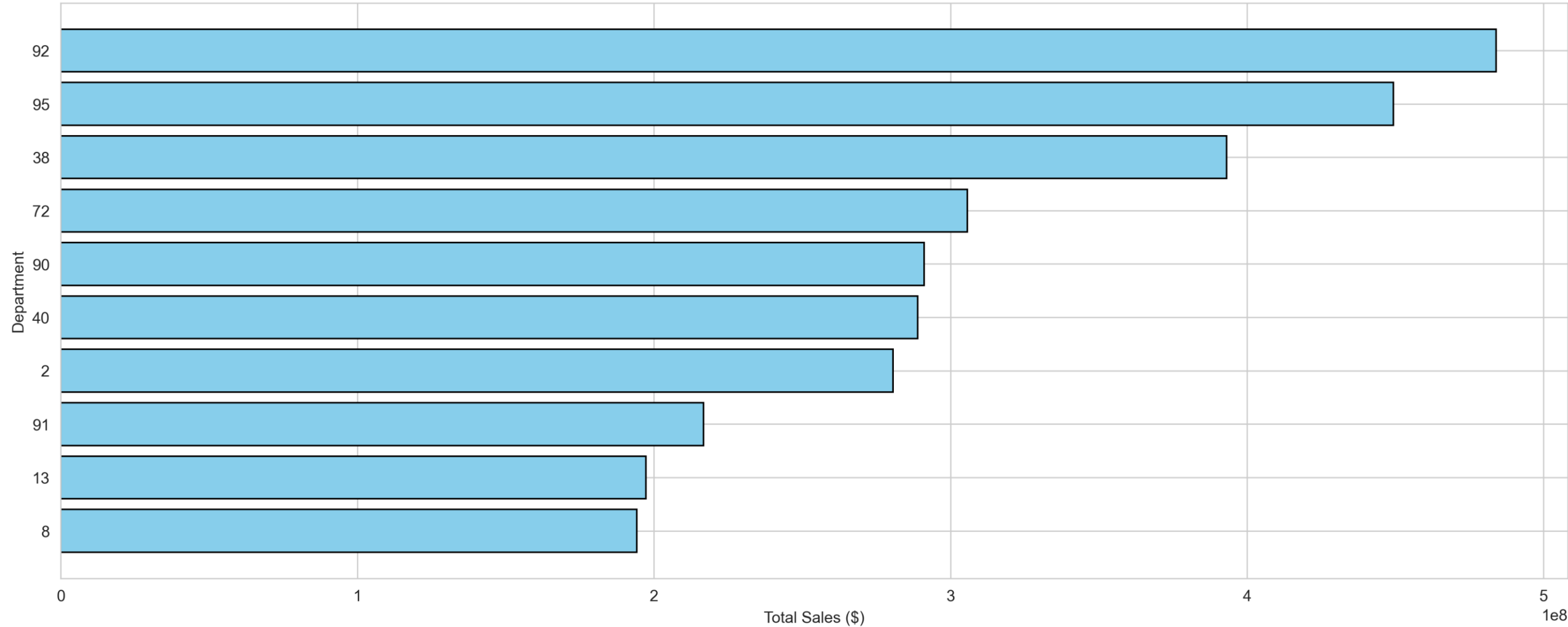


Step 1.4: EDA – Understanding the Data



Step 1.4: EDA – Understanding the Data

Top 10 Departments by Total Sales



Time Features (Step 1.3.1)

Week_Sin = $\sin(2\pi * \text{week} / \text{max_week})$

Week_Cos = $\cos(2\pi * \text{week} / \text{max_week})$

V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	
Year	Month	Day	Quarter	DayOfWeek	WeekOfYear	Is_Weekend	Is_Month_Start	Is_Month_End	Is_Quarter_Start	Is_Quarter_End	Is_Year_Start	Is_Year_End	Month_Sin	Month_Cos	Week_Sin	Week_Cos	DayOfWeek_Sin	DayOfWeek_Cos	
2010	2	5	1	4	5	0	0	0	0	0	0	0	0.866025	0.5	0.568064747	0.82298387	-0.433883739	-0.90096887	
2010	2	12	1	4	6	0	0	0	0	0	0	0	0.866025	0.5	0.663122658	0.74851075	-0.433883739	-0.90096887	
2010	2	19	1	4	7	0	0	0	0	0	0	0	0.866025	0.5	0.748510748	0.66312266	-0.433883739	-0.90096887	
2010	2	26	1	4	8	0	0	0	0	0	0	0	0.866025	0.5	0.822983866	0.56806475	-0.433883739	-0.90096887	
2010	3	5	1	4	9	0	0	0	0	0	0	0	1	6.12E-17	0.885456026	0.46472317	-0.433883739	-0.90096887	
2010	3	12	1	4	10	0	0	0	0	0	0	0	1	6.12E-17	0.935016243	0.35460489	-0.433883739	-0.90096887	
2010	3	19	1	4	11	0	0	0	0	0	0	0	1	6.12E-17	0.970941817	0.23931566	-0.433883739	-0.90096887	
2010	3	26	1	4	12	0	0	0	0	0	0	0	1	6.12E-17	0.992708874	0.12053668	-0.433883739	-0.90096887	
2010	4	2	2	4	13	0	0	0	0	0	0	0	0.866025	-0.5	1	-1.61E-16	-0.433883739	-0.90096887	
2010	4	9	2	4	14	0	0	0	0	0	0	0	0.866025	-0.5	0.992708874	-0.12053668	-0.433883739	-0.90096887	
2010	4	16	2	4	15	0	0	0	0	0	0	0	0.866025	-0.5	0.970941817	-0.23931566	-0.433883739	-0.90096887	
2010	4	23	2	4	16	0	0	0	0	0	0	0	0.866025	-0.5	0.935016243	-0.35460489	-0.433883739	-0.90096887	
2010	4	30	2	4	17	0	0	1	0	0	0	0	0.866025	-0.5	0.885456026	-0.46472317	-0.433883739	-0.90096887	
2010	5	7	2	4	18	0	0	0	0	0	0	0	0.5	-0.86603	0.822983866	-0.56806475	-0.433883739	-0.90096887	
2010	5	14	2	4	19	0	0	0	0	0	0	0	0.5	-0.86603	0.748510748	-0.66312266	-0.433883739	-0.90096887	
2010	5	21	2	4	20	0	0	0	0	0	0	0	0.5	-0.86603	0.663122658	-0.74851075	-0.433883739	-0.90096887	
2010	5	28	2	4	21	0	0	0	0	0	0	0	0.5	-0.86603	0.568064747	-0.82298387	-0.433883739	-0.90096887	

Lag & Rolling Features (Step 1.3.2)

Lag features give the model memory of previous weeks' sales.
Rolling features summarize recent trends and stability in sales.

AO	AP	AQ	AR	AS	AT	AU	AV
Sales_Lag1	Sales_Lag2	Sales_Lag4	Sales_Rolling_Mean_4	Sales_Rolling_Mean_8	Sales_Rolling_Std_4	Sales_Momentum	
0	0	0	24924.5	24924.5	0	0	
24924.5	0	0	35481.995	35481.995	14930.55261	21114.99	
46039.49	24924.5	0	37519.84667	37519.84667	11131.90096	-4443.94	
41595.55	46039.49	0	32990.77	32990.77	12832.10639	-22192	
19403.54	41595.55	24924.5	32216.62	30758.196	13554.04719	2424.36	
21827.9	19403.54	46039.49	25967.595	29139.06167	10467.48402	-784.51	
21043.39	21827.9	41595.55	21102.8675	28138.71571	1222.784968	1093.25	
22136.64	21043.39	19403.54	22809.285	27900.0275	2325.929203	4092.57	
26229.21	22136.64	21827.9	31666.9175	31941.76875	17206.39126	31029.22	
57258.43	26229.21	21043.39	37146.2975	31556.94625	16153.22682	-14297.5	
42960.91	57258.43	22136.64	36011.3775	28557.1225	17649.50249	-25364	
17596.96	42960.91	26229.21	33490.4125	28149.84875	20067.07091	-1451.61	
16145.35	17596.96	57258.43	23314.5825	27490.75	13111.79818	409.76	
16555.11	16145.35	42960.91	16927.84	27037.06875	691.6726194	858.83	
17413.94	16555.11	17596.96	17260.285	26635.83125	1230.316214	1512.8	
18926.74	17413.94	16145.35	16917.2075	25203.81	1733.352524	-4153.7	

Lag & Rolling Features (Step 1.3.2)

Our target from this is to conclude data that will help the prediction model

Lag = past value

Feature	Meaning
Sales_Lag1	sales last week
Sales_Lag2	sales 2 weeks ago
Sales_Lag4	sales 4 weeks ago

Rolling = average over several weeks

Feature	Meaning
Sales_Rolling_Mean_4	average sales of last 4 weeks
Sales_Rolling_Mean_8	average sales of last 8 weeks
Sales_Rolling_Std_4	how much sales vary in last 4 weeks
Sales_Momentum	is sales going up or down

Categorical Encoding (Step 1.3.3)

Actually , this part was in the row data, but it was represented using True/False

Store types were transformed using one-hot encoding.

This allows the model to learn differences between small, medium, and large stores.

B		AU	AV	AW
Type		Type_A	Type_B	Type_C
A		TRUE	FALSE	FALSE
A		TRUE	FALSE	FALSE
B		TRUE	FALSE	FALSE
A		TRUE	FALSE	FALSE
B		TRUE	FALSE	FALSE
A		TRUE	FALSE	FALSE
B		TRUE	FALSE	FALSE
A		TRUE	FALSE	FALSE
B		TRUE	FALSE	FALSE
B		TRUE	FALSE	FALSE

Feature Normalization (Step 1.3.4)

We standardized 17 numeric features using z-score normalization.

To avoid data leakage, we saved normalization parameters separately for consistent use during inference.

Why: to not make the model think one value is more important than another.

$$\text{Normalized Value} = \frac{x - \mu}{\sigma}$$

Where x refers to the original value

μ is the mean of the data

σ is the standard deviation of data

Feature	Original Value	After Normalization
Weekly_Sales	24,924	-0.70
Temperature	12.5	-1.17
CPI	211.3	1.02
Fuel_Price	3.10	-1.72

```
{
  "Size": {
    "mean": 136727.91573878596,
    "std": 60980.583328101486
  },
  "Temperature": {
    "mean": 60.09005873283203,
    "std": 18.44793114761593
  },
  "Fuel_Price": {
    "mean": 3.361026527029912,
    "std": 0.4585145371277508
  },
  "CPI": {
    "mean": 171.20194682296346,
    "std": 39.1592756230124
  },
  "Unemployment": {
    "mean": 7.96028869464146,
    "std": 1.8632960384301736
  },
}
```

Final Dataset Summary

At the end of Stage 1, both training and test data were fully cleaned, transformed, and enriched.

This dataset is now suitable for machine learning and deep learning forecasting models.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P
1	Store	Dept	Date	Weekly_Sales	Holiday	Type	Size	Temperature	Fuel_Price	MarkDown	MarkDown	MarkDown	MarkDown	MarkDown	CPI	Unemployment
2	1	1	5/02/2010	24924.5	FALSE	A	151315	42.31	2.572						211.0964	8.106
3	1	1	12/02/2010	46039.49	TRUE	A	151315	38.51	2.548						211.2422	8.106
4	1	1	19/02/2010	41595.55	FALSE	A	151315	39.93	2.514						211.2891	8.106
5	1	1	26/02/2010	19403.54	FALSE	A	151315	46.63	2.561						211.3196	8.106
6	1	1	5/03/2010	21827.9	FALSE	A	151315	46.5	2.625						211.3501	8.106
7	1	1	12/03/2010	21043.39	FALSE	A	151315	57.79	2.667						211.3806	8.106
8	1	1	19/03/2010	22136.64	FALSE	A	151315	54.58	2.72						211.2156	8.106
9	1	1	26/03/2010	26229.21	FALSE	A	151315	51.45	2.732						211.018	8.106
10	1	1	2/04/2010	57258.43	FALSE	A	151315	62.27	2.719						210.8204	7.808
11	1	1	9/04/2010	42960.91	FALSE	A	151315	65.86	2.77						210.6229	7.808
12	1	1	16/04/2010	17596.96	FALSE	A	151315	66.32	2.808						210.4887	7.808
13	1	1	23/04/2010	16145.35	FALSE	A	151315	64.84	2.795						210.4391	7.808
14	1	1	30/04/2010	16555.11	FALSE	A	151315	67.41	2.78						210.3895	7.808
15	1	1	7/05/2010	17413.94	FALSE	A	151315	72.55	2.835						210.34	7.808
16	1	1	14/05/2010	18926.74	FALSE	A	151315	74.78	2.854						210.3374	7.808
17	1	1	21/05/2010	14773.04	FALSE	A	151315	76.44	2.826						210.6171	7.808
18	1	1	28/05/2010	15580.43	FALSE	A	151315	80.44	2.759						210.8968	7.808
19	1	1	4/06/2010	17558.09	FALSE	A	151315	80.69	2.705						211.1764	7.808
20	1	1	11/06/2010	16637.62	FALSE	A	151315	80.43	2.668						211.4561	7.808
21	1	1	18/06/2010	16216.27	FALSE	A	151315	84.11	2.637						211.4538	7.808
22	1	1	25/06/2010	16328.72	FALSE	A	151315	84.34	2.653						211.3387	7.808
23	1	1	2/07/2010	16333.14	FALSE	A	151315	80.91	2.669						211.2235	7.787
24	1	1	9/07/2010	17688.76	FALSE	A	151315	80.48	2.642						211.1084	7.787
25	1	1	16/07/2010	17150.84	FALSE	A	151315	83.15	2.623						211.1004	7.787
26	1	1	23/07/2010	15360.45	FALSE	A	151315	83.36	2.608						211.2351	7.787

train.csv Before

Final Dataset Summary

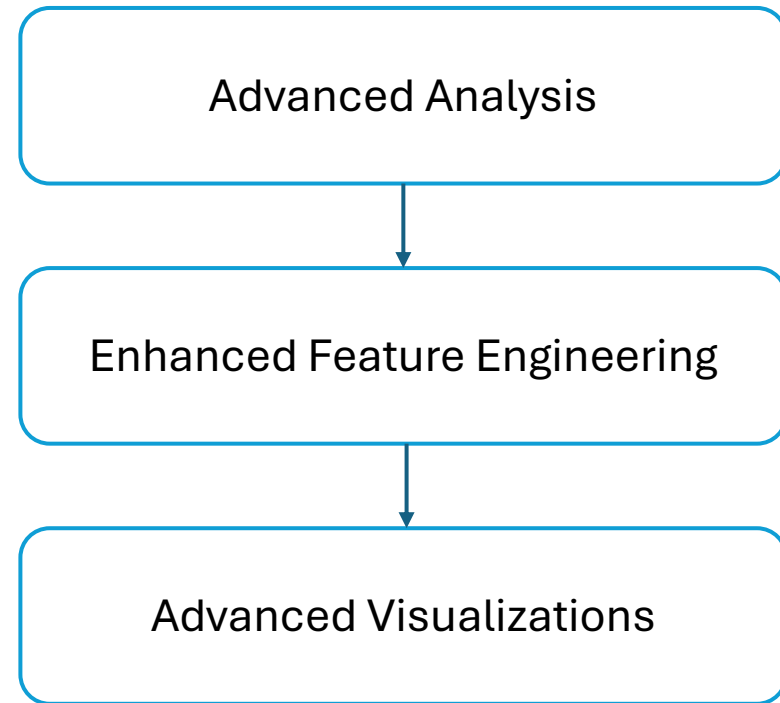
train_final.csv After

A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT	AU	AV	AW		
1	Dept	Date	Weekly	IsHolid	Size	Temper	Fuel_Pri	MarkDo	MarkDo	MarkDo	MarkDo	CPI	Unemplc	Has_Ms	Has_Ms	Has_Ms	Has_Ms	Has_Ms	Year	Month	Day	Quarter	DayOfW	WeekOf	Is_Week	Is_Mont	Is_Mont	Is_Quart	Is_Quart	Is_Year	Is_Year	Month_1	Month_1	Week_S	Week_C	DayOfW	DayOfW	Sales_L	Sales_L	Sales_L	Sales_R	Sales_R	Sales_R	Sales_R	Sales_N	Type_A	Type_B	Type_C		
2	1	####	24925	FALSE	0.2392	-0.9638	-1.7208	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0188	0.0782	0	0	0	0	0	2010	2	5	1	4	5	0	0	0	0	0	0	0	0.866	0.5	0.5681	0.823	-0.4339	-0.901	-0.6998	-0.6961	-0.6887	0.4013	0.4035	-0.3826	0.0014	TRUE	FALSE	FALSE		
3	1	####	46039	TRUE	0.2392	-1.1698	-1.7732	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0225	0.0782	0	0	0	0	0	2010	2	12	1	4	6	0	0	0	0	0	0	0	0.866	0.5	0.6631	0.7485	-0.4339	-0.901	0.3991	-0.6961	-0.6887	0.8758	0.8812	2.5519	2.9429	TRUE	FALSE	FALSE		
4	1	####	41596	FALSE	0.2392	-1.0928	-1.8473	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0237	0.0782	0	0	0	0	0	2010	2	19	1	4	7	0	0	0	0	0	0	0	0.866	0.5	0.7485	0.6631	-0.4339	-0.901	1.33	0.4044	-0.6887	0.9674	0.9735	1.8053	-0.6177	TRUE	FALSE	FALSE		
5	1	####	19404	FALSE	0.2392	-0.7296	-1.7448	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0245	0.0782	0	0	0	0	0	2010	2	26	1	4	8	0	0	0	0	0	0	0	0.866	0.5	0.823	0.5681	-0.4339	-0.901	1.1341	1.3366	-0.6887	0.7638	0.7685	2.1395	-3.0901	TRUE	FALSE	FALSE		
6	1	####	21828	FALSE	0.2392	-0.7367	-1.6052	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0253	0.0782	0	0	0	0	0	2010	3	5	1	4	9	0	0	0	0	0	0	0	1	6.12E-17	0.8855	0.4647	-0.4339	-0.901	0.1556	1.1404	0.4156	0.729	0.6675	2.2814	0.3391	TRUE	FALSE	FALSE		
7	1	####	21043	FALSE	0.2392	-0.1247	-1.5136	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.026	0.0782	0	0	0	0	0	2010	3	12	1	4	10	0	0	0	0	0	0	0	1	6.12E-17	0.935	0.3546	-0.4339	-0.901	0.2625	0.1606	1.3512	0.4482	0.5942	1.6747	-0.1079	TRUE	FALSE	FALSE		
8	1	####	22137	FALSE	0.2392	-0.2987	-1.3981	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0218	0.0782	0	0	0	0	0	2010	3	19	1	4	11	0	0	0	0	0	0	0	1	6.12E-17	0.9709	0.2393	-0.4339	-0.901	0.2279	0.2676	1.1543	0.2296	0.549	-0.1423	0.1537	TRUE	FALSE	FALSE		
9	1	####	26229	FALSE	0.2392	-0.4683	-1.3719	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0168	0.0782	0	0	0	0	0	2010	3	26	1	4	12	0	0	0	0	0	0	0	1	6.12E-17	0.9927	0.1205	-0.4339	-0.901	0.2761	0.233	0.171	0.3062	0.5382	0.0745	0.5715	TRUE	FALSE	FALSE		
10	1	####	57258	FALSE	0.2392	0.1182	-1.4002	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0117	-0.0817	0	0	0	0	0	2010	4	2	2	4	13	0	0	0	0	0	0	0	0.866	-0.5	1	-1.61E-16	-0.4339	-0.901	0.4566	0.2813	0.2784	0.7043	0.721	2.9993	4.324	TRUE	FALSE	FALSE		
11	1	####	42961	FALSE	0.2392	0.3128	-1.289	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0067	-0.0817	0	0	0	0	0	2010	4	9	2	4	14	0	0	0	0	0	0	0	0.866	-0.5	0.9927	-0.1205	-0.4339	-0.901	0.18246	0.462	0.2437	0.9506	0.7036	2.7923	-1.9904	TRUE	FALSE	FALSE		
12	1	####	17597	FALSE	0.2392	0.3377	-1.2061	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0033	-0.0817	0	0	0	0	0	2010	4	16	2	4	15	0	0	0	0	0	0	0	0.866	-0.5	0.9709	-0.2393	-0.4339	-0.901	1.1943	1.8319	0.2321	0.8996	0.5679	3.0863	-3.532	TRUE	FALSE	FALSE		
13	1	####	16145	FALSE	0.2392	0.2575	-1.2345	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.002	-0.0817	0	0	0	0	0	2010	4	23	2	4	16	0	0	0	0	0	0	0	0.866	-0.5	0.935	-0.3546	-0.4339	-0.901	0.076	1.2007	0.4734	0.7863	0.5495	3.5615	-2.0008	TRUE	FALSE	FALSE		
14	1	####	16555	FALSE	0.2392	0.3968	-1.2672	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0007	-0.0817	0	0	0	0	0	2010	4	30	2	4	17	0	0	0	1	0	0	0	0.866	-0.5	0.8855	-0.4647	-0.4339	-0.901	0.012	0.0908	1.8482	0.329	0.5196	2.1945	0.0585	TRUE	FALSE	FALSE		
15	1	####	17414	FALSE	0.2392	0.6754	-1.1472	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	0.9995	-0.0817	0	0	0	0	0	2010	5	7	2	4	18	0	0	0	0	0	0	0	0	0.5	-0.866	0.823	-0.5681	-0.4339	-0.901	0.0301	0.0167	1.2148	0.0419	0.9391	-0.2467	0.121	TRUE	FALSE	FALSE	
16	1	####	19327	FALSE	0.2392	0.7963	-1.1058	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	0.9994	-0.0817	0	0	0	0	0	2010	5	14	2	4	19	0	0	0	0	0	0	0	0	0.5	-0.866	0.7485	-0.6631	-0.4339	-0.901	0.0679	0.0348	0.091	0.0569	0.481	-0.1408	0.2121	TRUE	FALSE	FALSE	
17	1	####	14773	FALSE	0.2392	0.8963	-1.1663	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0065	-0.0817	0	0	0	0	0	2010	5	21	2	4	20	0	0	0	0	0	0	0	0	0.5	-0.866	0.6631	-0.7485	-0.4339	-0.901	0.1046	0.0728	0.0267	0.0414	0.0162	-0.0419	-0.5772	TRUE	FALSE	FALSE	
18	1	####	15580	FALSE	0.2392	1.1031	-1.313	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0137	-0.0817	0	0	0	0	0	2010	5	28	2	4	21	0	0	0	0	0	0	0	0	0.5	-0.866	0.5681	-0.823	-0.4339	-0.901	-0.0485	0.1395	0.0448	0.3065	0.1804	-0.0161	0.1139	TRUE	FALSE	FALSE	
19	1	####	17558	FALSE	0.2392	1.1167	-1.4308	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0208	-0.0817	0	0	0	0	0	2010	6	4	2	4	22	0	0	0	0	0	0	0	0	1.22E-16	-1	0.4647	-0.8855	-0.4339	-0.901	-0.0129	-0.0438	0.0829	0.0321	0.0367	-0.0121	0.2769	TRUE	FALSE	FALSE	
20	1	####	16638	FALSE	0.2392	1.1026	-1.5115	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.028	-0.0817	0	0	0	0	0	2010	6	11	2	4	23	0	0	0	0	0	0	0	0	1.22E-16	-1	0.3546	-0.935	-0.4339	-0.901	0.0743	-0.0082	0.1499	0.0064	0.0313	-0.1435	-0.1268	TRUE	FALSE	FALSE	
21	1	####	16216	FALSE	0.2392	1.302	-1.5791	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0279	-0.0817	0	0	0	0	0	2010	6	18	2	4	24	0	0	0	0	0	0	0	0	1.22E-16	-1	0.2393	-0.9709	-0.4339	-0.901	0.0337	0.0791	-0.0341	0.0226	0.0317	-0.2196	-0.0573	TRUE	FALSE	FALSE	
22	1	####	16329	FALSE	0.2392	1.3145	-1.5442	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.025	-0.0817	0	0	0	0	0	2010	6	25	2	4	25	0	0	0	0	0	0	0	0	1.22E-16	-1	0.1205	-0.9927	-0.4339	-0.901	0.0151	0.0385	0.0016	0.031	0.0304	-0.263	0.0171	TRUE	FALSE	FALSE	
23	1	####	16333	FALSE	0.2392	1.1286	-1.5093	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.022	-0.093	0	0	0	0	0	2010	7	2	3	4	26	0	0	0	0	0	0	0	0	0	-0.5	-0.866	#####	-1	-0.4339	-0.901	0.0201	0.0199	0.0892	0.0173	0.0243	-0.3471	0.002	TRUE	FALSE	FALSE
24	1	####	17689	FALSE	0.2392	1.1053	-1.5682	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0191	-0.093	0	0	0	0	0	2010	7	9	3	4	27	0	0	0	0	0	0	0	0	0	-0.5	-0.866	-0.1205	-0.9927	-0.4339	-0.901	0.0203	0.0248	0.0485	0.0291	0.0173	-0.245	0.1902	TRUE	FALSE	FALSE
25	1	####	17151	FALSE	0.2392	1.25	-1.6096	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0189	-0.093	0	0	0	0	0	2010	7	16	3	4	28	0	0	0	0	0	0	0	0	0	-0.5	-0.866	-0.2393	-0.9709	-0.4339	-0.901	0.08	0.025	0.0298	0.0396	0.0308	-0.2517	-0.0735	TRUE	FALSE	FALSE
26	1	####	15360	FALSE	0.2392	1.2614	-1.6423	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0223	-0.093	0	0	0	0	0	2010	7	23	3	4	29	0	0	0	0	0	0	0	0	0	-0.5	-0.866	-0.3546	-0.935	-0.4339	-0.901	0.0563	0.0849	0.0348	0.0287	0.0295	-0.1831	-0.248	TRUE	FALSE	FALSE
27	1	####	15382	FALSE	0.2392	1.179	-1.5725	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0258	-0.093	0	0	0	0	0	2010	7	30	3	4	30	0	0	0	0	0	0	0	0	0	-0.5	-0.866	-0.4647	-0.8855	-0.4339	-0.901	-0.0226	0.0611	0.035	0.018	0.0172	-0.1462	0.0044	TRUE	FALSE	FALSE
28	1	####	17508	FALSE	0.2392	1.4674	-1.6009	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0292	-0.093	0	0	0	0	0	2010	8	6	3	4	31	0	0	0	0	0	0	0	0	0	-0.866	-0.5	-0.5681	-0.823	-0.4339	-0.901	-0.0217	-0.0179	0.095	0.016	0.0021	-0.1585	0.2977	TRUE	FALSE	FALSE
29	1	####	15536	FALSE	0.2392	1.4587	-1.4591	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1.0326	-0.093	0	0	0	0	0	2010	8	13	3	4	32	0	0	0	0	0	0	0	0	0	-0.866	-0.5	-0.6631	-0.7485	-0.4339	-										

Milestone 2: Advanced Data Analysis & Feature Engineering

In Milestone 2 we performed advanced time-series analysis and expanded our feature set to improve forecasting accuracy.

Stage 2 Overview



Task 2.1: Advanced Data Analysis (overview)

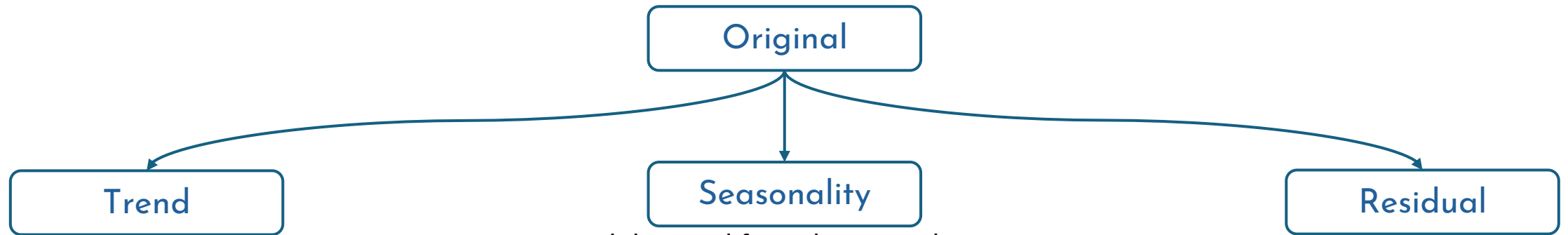
Task 2.1 performed time-series decomposition, stationarity testing, correlation analysis, and quantified holiday effects.

- Time Series Decomposition (Trend, Seasonality, Residual)
- ADF Stationarity Test
- Correlation Analysis (heatmap + top predictors)
- Holiday Impact Analysis

Time Series Decomposition (Trend, Seasonality, Residual)

First, we aggregated the data into a single weekly time series by summing weekly sales across all stores and departments, and sorted it by date.

```
weekly_sales = train.groupby('Date')['Weekly_Sales'].sum()
```



Then, we extracted the trend using a 52-week rolling mean to capture the long-term sales movement.

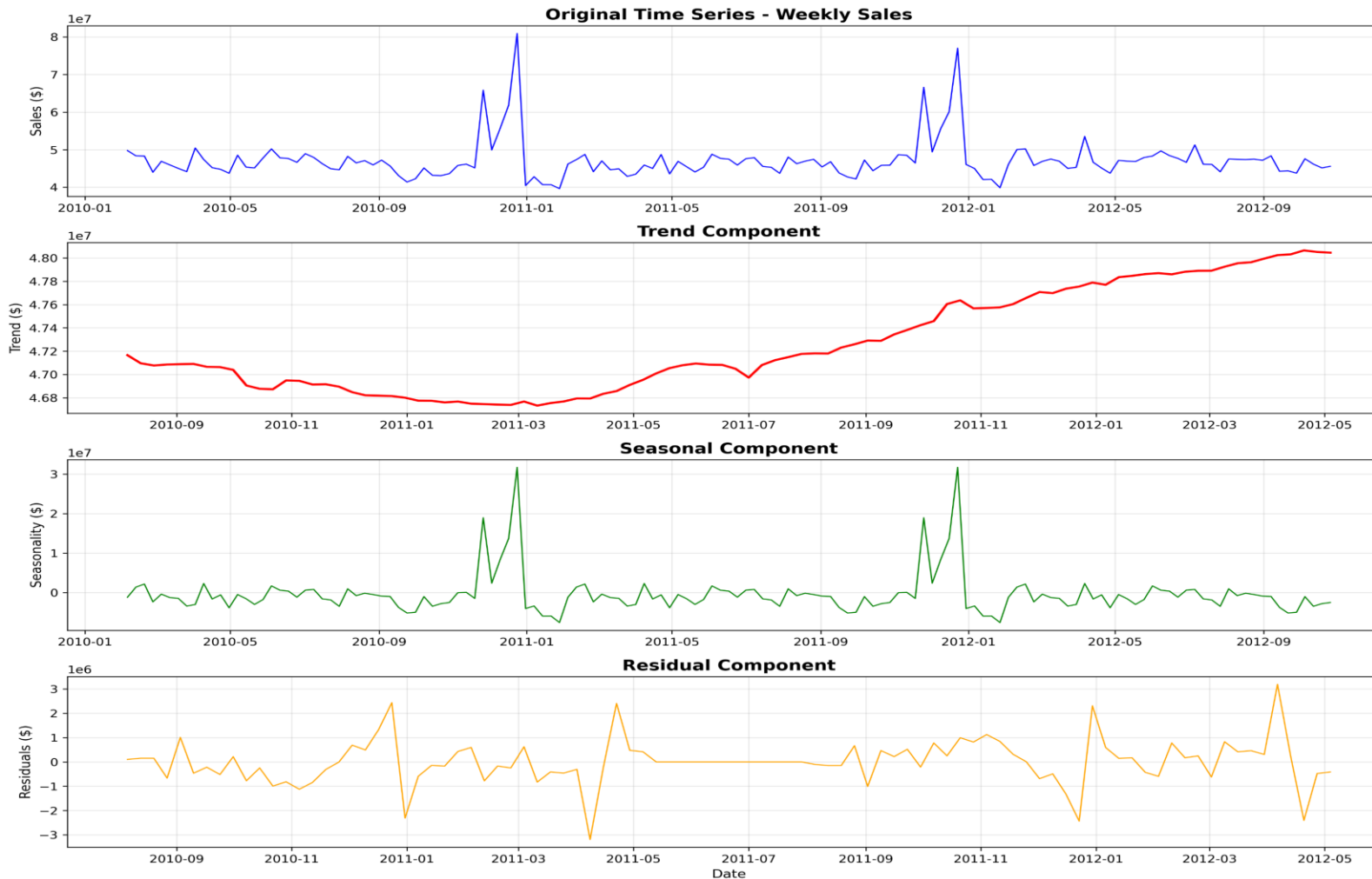
```
trend =  
weekly_sales.rolling(window=52,  
center=True).mean()
```

removed the trend from the original series to isolate seasonal effects.
Then calculated seasonality by averaging the **detrended** values for each week of the year, capturing repeating annual patterns.

$\text{residual} = \text{detrended} - \text{seasonality}$

Task 2.1: Advanced Data Analysis (overview)

(trend / seasonality / residuals)



Stationarity (ADF results)

ADF tests show the sales series is non-stationary ($p\text{-value} > 0.05$), so differencing is required for ARIMA-type models to make the series stationary.

ADF basically answers:

"Does this data have a trend or not?"

The p-value tells you if the ADF test result is significant or not.

$p\text{-value} > 0.05 \rightarrow$ the sales data is non-stationary.

```
1  {
2    "original_mean": 47113419.49027973,
3    "original_std": 5444206.20253656,
4    "differenced_mean": -29624.114154929583,
5    "differenced_std": 6284025.500197348,
6    "variance_ratio": 1.1542592742481907,
7    "mean_stability_ratio": 0.004714193816368066,
8    "is_stationary_str": "No",
9    "interpretation": "Series appears to be NON-STATIONARY (requires differencing)"
10 }
```

What Is ARIMA (as we are preparing for forecasting)

A = Auto

R = Regressive (uses past values)

I = Integrated (this means differencing to remove trends)

MA = Moving Average (uses past errors)

And these models **require the data to be stationary.**

So if ADF shows non-stationarity ($p > 0.05$), we must use: **Differencing**

Example:

$$y_{\text{diff}} = y(t) - y(t-1)$$

This removes the trend and makes the series stationary.

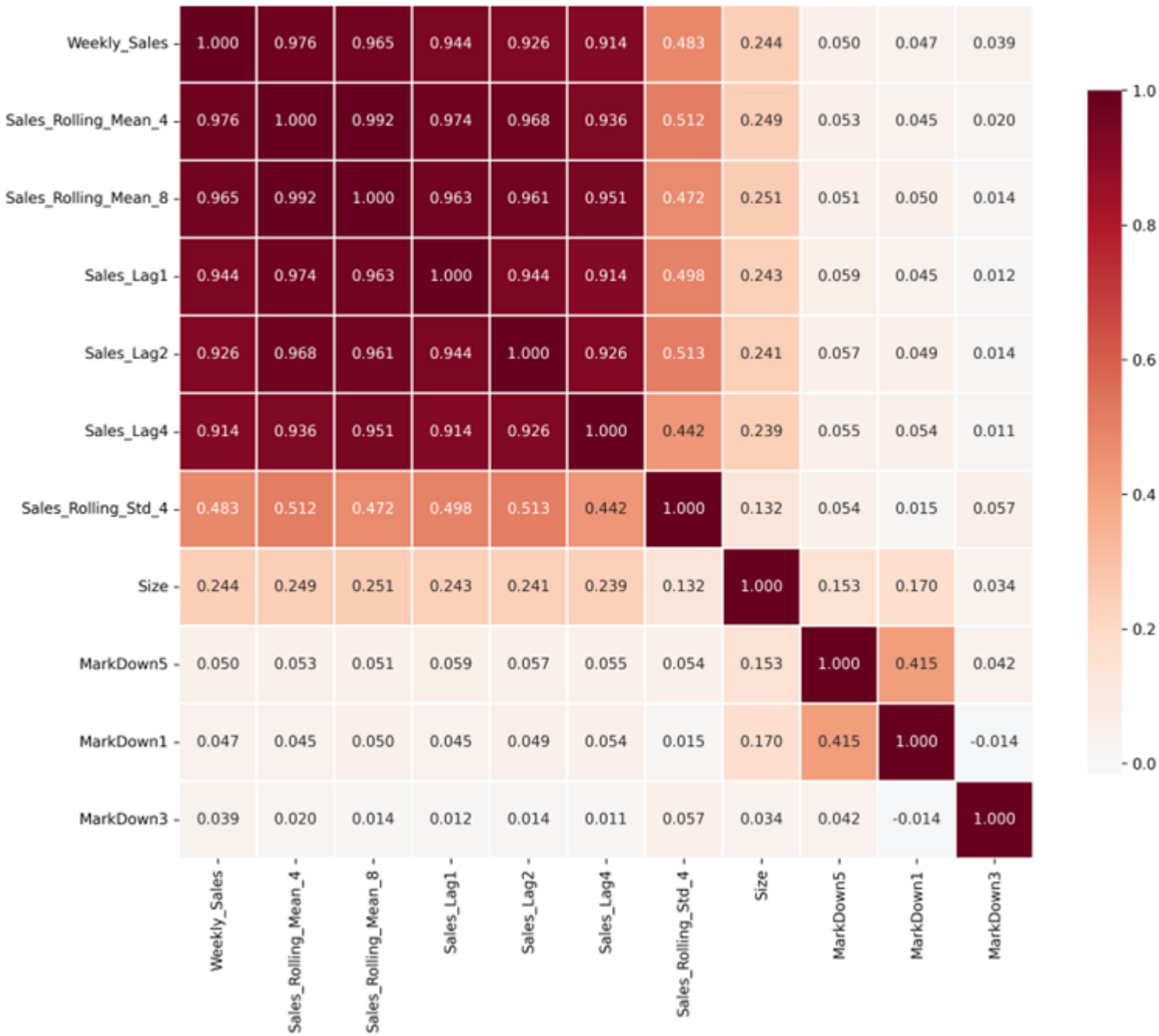
Correlation Analysis (heatmap)

Correlation analysis confirms that lag and rolling features dominate; the heatmap highlights which engineered features align most strongly with Weekly_Sales.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S
1		Weekly_Sales	Temperature	Fuel_Price	CPI	Unemployment	Size	MarkDown1	MarkDown2	MarkDown3	MarkDown4	MarkDown5	Sales_Lag1	Sales_Lag2	Sales_Lag4	Sales_Rolling_Mean_4	Sales_Rolling_Mean_8	Sales_Rolling_Std_4	
2	Weekly_Sales	1	-0.002312447	-0.000120296	-0.02092	-0.025863716	0.243828	0.047172342	0.02071647	0.038562	0.03746718	0.05046529	0.94377404	0.925972305	0.9135021	0.975797193	0.964755853	0.483369301	
3	Temperature	-0.00231245	1	0.143858819	0.182112	0.09673018	-0.05831	-0.026415133	-0.1796724	-0.0560263	-0.0502809	-0.0147524	0.00335106	0.009995424	0.0213063	-0.004321042	-0.005316009	-0.11208593	
4	Fuel_Price	-0.0001203	0.143858819	1	-0.16421	-0.033852703	0.003361	0.297056262	0.02915324	0.0186149	0.16662195	0.2154197	0.00621265	0.013718487	0.0286293	-0.00068927	-0.000511022	-0.01644422	
5	CPI	-0.02092134	0.182111607	-0.164210406	1	-0.299953018	-0.00331	0.010914758	-0.0035539	-0.0058388	-0.0020471	0.0679059	-0.0202559	-0.01966489	-0.0184948	-0.021359752	-0.021522366	-0.0286378	
6	Unemployment	-0.02586372	0.09673018	-0.033852703	-0.29995	1	-0.06824	-0.105168294	-0.0414273	-0.0180776	-0.0765128	-0.1204063	-0.0272485	-0.02959018	-0.0333179	-0.026270532	-0.026595399	-0.00492487	
7	Size	0.243827992	-0.05831294	0.00336119	-0.00331	-0.068238254	1	0.16978755	0.0783722	0.0336414	0.12733356	0.1530107	0.24251701	0.24127981	0.2387273	0.249194062	0.251137919	0.132097449	
8	MarkDown1	0.047172342	-0.026415133	0.297056262	0.010915	-0.105168294	0.169788	1	0.17486768	-0.0144109	0.83890442	0.41505018	0.04542355	0.049188963	0.0535339	0.045273735	0.049720778	0.01545521	
9	MarkDown2	0.020716467	-0.179672448	0.029153237	-0.00355	-0.041427281	0.078372	0.174867684	1	-0.0060802	0.11324999	0.13173509	0.05985109	0.058043874	0.038626	0.046607267	0.043366459	0.110966022	
10	MarkDown3	0.038562039	-0.056026324	0.018614902	-0.00584	-0.018077638	0.033641	-0.01441094	-0.0060802	1	-0.0120201	0.04247116	0.01241732	0.013762764	0.011278	0.019500461	0.014247461	0.056733123	
11	MarkDown4	0.037467181	-0.05028091	0.166621948	-0.00205	-0.076512836	0.127334	0.838904415	0.11324999	-0.0120201	1	0.30336958	0.03123029	0.033545636	0.0390108	0.032403948	0.039949736	0.006785411	
12	MarkDown5	0.050465293	-0.014752445	0.215419702	0.067906	-0.120406278	0.153011	0.415050183	0.13173509	0.0424712	0.30336958	1	0.05947685	0.056881829	0.0554456	0.053382071	0.051162905	0.054057422	
13	Sales_Lag1	0.943774037	0.003351063	0.006212649	-0.02026	-0.027248469	0.242517	0.045423545	0.05985109	0.0124173	0.03123029	0.05947685	1	0.943691344	0.9135993	0.974350531	0.963447666	0.497520254	
14	Sales_Lag2	0.925972305	0.009995424	0.013718487	-0.01966	-0.029590185	0.24128	0.049188963	0.05804387	0.0137628	0.03354564	0.05688183	0.94369134	1	0.9257017	0.968490308	0.960903445	0.513473823	
15	Sales_Lag4	0.913502079	0.021306341	0.028629285	-0.01849	-0.033317855	0.238727	0.053533932	0.03862599	0.011278	0.0390108	0.05544561	0.91359925	0.925701744	1	0.935767766	0.951454402	0.441709021	
16	Sales_Rolling_Mean_4	0.975797193	-0.004321042	-0.00068927	-0.02136	-0.026270532	0.249194	0.045273735	0.04660727	0.0195005	0.03240395	0.05338207	0.97435053	0.968490308	0.9357678	1	0.992320955	0.512224513	
17	Sales_Rolling_Mean_8	0.964755853	-0.005316009	-0.000511022	-0.02152	-0.026595399	0.251138	0.049720778	0.04336646	0.0142475	0.03994974	0.05116291	0.96344767	0.960903445	0.9514544	0.992320955	1	0.472473955	
18	Sales_Rolling_Std_4	0.483369301	-0.112085928	-0.016444222	-0.02864	-0.00492487	0.132097	0.01545521	0.11096602	0.0567331	0.00678541	0.05405742	0.49752025	0.513473823	0.441709	0.512224513	0.472473955	1	
19																			
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correlation_matrix

Correlation Heatmap - Top Features vs Weekly Sales

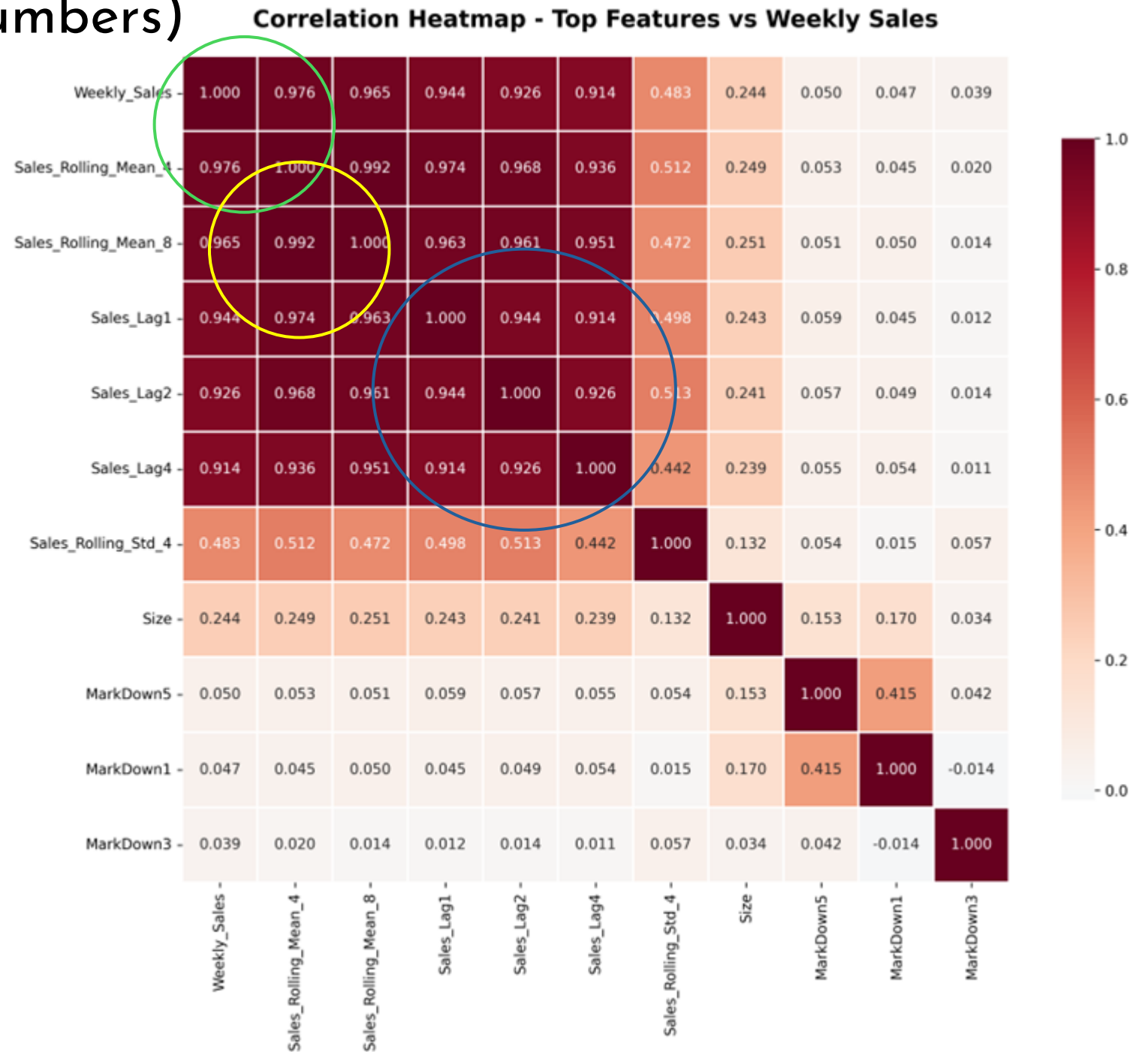


Top Correlated Features (numbers)

The 4-week rolling average has the strongest correlation with weekly sales. This tells us that recent short-term trends are the most powerful predictor of next week's sales.

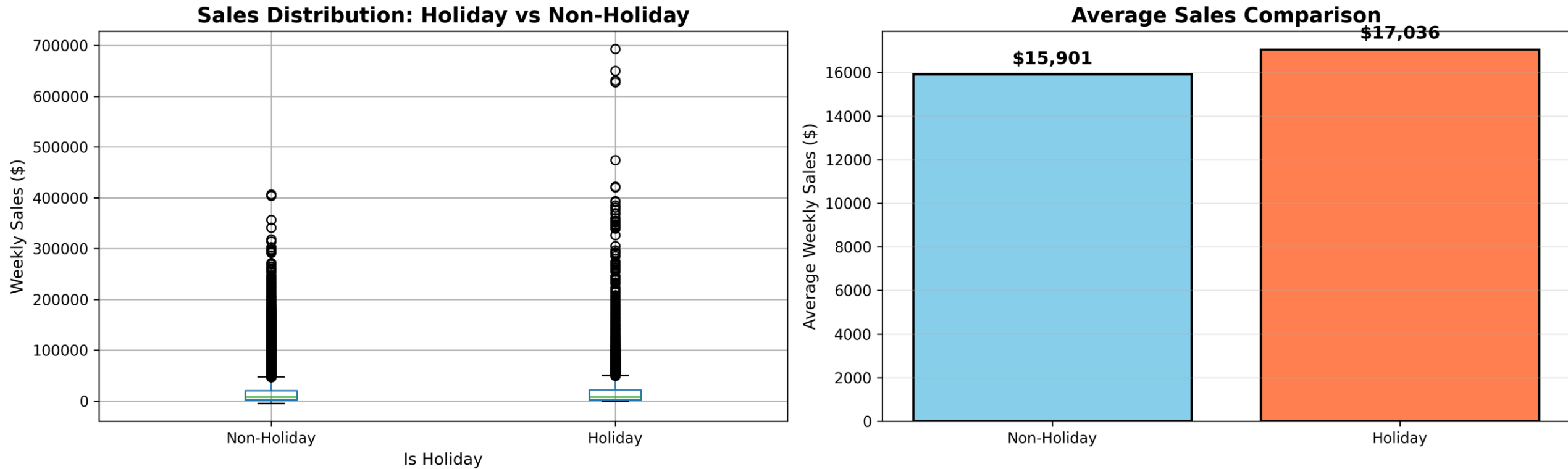
The 8-week rolling average is also extremely strong. This means the model benefits from understanding medium-term sales patterns, not just the last few weeks.

Recent past sales, especially 1-week, 2-week, and 4-week lags, all show very strong correlations, confirming that weekly sales strongly depend on short-term history and recurring monthly patterns.



Holiday Impact Analysis

Holiday Impact on Sales



Holidays increase average sales by +7.13% and increase volatility by +21.9%. We therefore created specific holiday-season and 'days-to-holiday' features.

Task 2.2: Enhanced Feature Engineering (overview)

We added 42 new advanced features across 6 categories:

1- Advanced Rolling Statistics

EMAs, rolling min/max, ranges, trends, coefficient of variation, acceleration.

2- Seasonal & Holiday Features

Holiday season flags, back-to-school, Super Bowl, days until Thanksgiving/Christmas, seasonal dummies.

3- Store Performance Features

Store averages, department averages, store-dept combos, deviations from averages.

4- Promotional Features

Total markdowns, number of active promotions, promo intensity, rolling promo averages.

5- Economic Interaction Features

CPI × unemployment, holiday temperature, fuel price × unemployment, purchasing power.

6- Time-Based Aggregations

Monthly and quarterly average sales, total sales, YoY store growth.

Example: Advanced Rolling Stats (EMAs & Acceleration)

AX	AY	BA	BB	BC	BD	BF
Sales_EMA_4	Sales_EMA_8	Sales_Rolling_Min_4	Sales_Rolling_Max_4	Sales_Rolling_Range_4	Sales_Trend	Sales_Acceleration
24924.5	24924.5	24924.5	24924.5	0	0	
33370.496	29616.72	24924.5	46039.49	21114.99	5197.536	2.941501655
36660.5176	32278.68222	24924.5	46039.49	21114.99	6422.543754	-3.560581127
29757.72656	29417.53951	19403.54	46039.49	26635.95	1186.58869	-2.472460431

- EMA → recent trend
- Min/Max/Range → volatility
- Trend → short-term direction
- Acceleration → how fast trend changes

EMAs give weighted recent trends, range/min/max capture extremes, and acceleration quantifies the speed of change – all crucial for short-term forecasting.

Example: Seasonal & Holiday Features

BG	BH	BI	BJ	BK	BL	BM	BN	BO
Is_Holiday_Season	Is_BackToSchool_Season	Is_SuperBowl_Week	Days_To_Thanksgiving	Days_To_Christmas	Season_Winter	Season_Spring	Season_Summer	Season_Fall
0	0	1	292	323	1	0	0	0
0	0	1	285	316	1	0	0	0
0	0	0	278	309	1	0	0	0
0	0	0	271	302	1	0	0	0
0	0	0	264	295	0	1	0	0
0	0	0	257	288	0	1	0	0
0	0	0	250	281	0	1	0	0

These features explicitly encode holiday timing and season boundaries so models can react to predictable calendar-driven spikes.

Example: Store/Dept Aggregates & Deviations

BP	BQ	BR	BS	BT	BU	BV
Store_Avg_Sales	Store_Std_	Store_Min_	Store_Max_	Dept_Avg_Sales	Dept_Std_	StoreDept_Avg_Sales
21710.54362	27748.95	-863	203670.5	19213.48509	15102.4	22513.32294
21710.54362	27748.95	-863	203670.5	19213.48509	15102.4	22513.32294
21710.54362	27748.95	-863	203670.5	19213.48509	15102.4	22513.32294
21710.54362	27748.95	-863	203670.5	19213.48509	15102.4	22513.32294
21710.54362	27748.95	-863	203670.5	19213.48509	15102.4	22513.32294

Store	Dept	Weekly_Sales	Store_Avg_Sales	Dept_Avg_Sales	StoreDept_Avg_Sales
1	1	24924.5	21710.5	19213.5	22513.3

The dataset contains many stores and many departments. So we created summary features that describe how each store, each department, and each store-department combination usually performs. These help the model understand the baseline sales and whether a particular week is above or below normal.

Task 2.3: Advanced Visualizations (dashboard)

We produced a 7-panel dashboard and multiple plots that helped validate feature choices and communicated the patterns to stakeholders.

WALMART SALES FORECASTING - COMPREHENSIVE DASHBOARD



Task 2.3: Final dataset summary

train_enhanced.csv before

	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	AA	AB	AC	AD	AE	AF	AG	AH	AI	AJ	AK	AL	AM	AN	AO	AP	AQ	AR	AS	AT	AU	AV	AW			
Dept	Date	Weekly	IsHolid	Size	Temper	Fuel_Pri	MarkDo	MarkDo	MarkDo	MarkDo	CPI	Unemplc	Has_Mz	Has_Mz	Has_Mz	Has_Mz	Has_Mz	Has_Mz	Year	Month	Day	Quarter	DayOfW	WeekOf	Is_Week	Is_Mont	Is_Mont	Is_Quart	Is_Quart	Is_Year	Is_Year	Month	Month	Week	Week	Week	DayOfW	DayOfW	Sales_L	Sales_L	Sales_L	Sales_R	Sales_R	Sales_R	Sales_R	Sales_R	Type_A	Type_B	Type_C		
1	#####	24325	FALSE	0.2392	-0.9638	-1.7208	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10188	0.0782	0	0	0	0	0	0	2010	2	5	1	4	5	0	0	0	0	0	0	0	0.866	0.5	0.5681	0.823	-0.4339	-0.901	-0.6398	-0.6961	-0.6887	0.4013	0.4035	-0.3826	0.0014	TRUE	FALSE	FALSE			
1	#####	46039	TRUE	0.2392	-1.1638	-1.7732	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10225	0.0782	0	0	0	0	0	0	2010	2	12	1	4	6	0	0	0	0	0	0	0	0.866	0.5	0.6631	0.7485	-0.4339	-0.901	0.3391	-0.6361	-0.6887	0.8758	0.8812	2.5519	2.9429	TRUE	FALSE	FALSE			
1	#####	41596	FALSE	0.2392	-1.0928	-1.8473	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10237	0.0782	0	0	0	0	0	0	2010	2	19	1	4	7	0	0	0	0	0	0	0	0.866	0.5	0.7485	0.6631	-0.4339	-0.901	1.33	0.4044	-0.6887	0.9674	0.9735	1.8053	-0.1677	TRUE	FALSE	FALSE			
1	#####	19404	FALSE	0.2392	-0.7236	-1.7448	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10245	0.0782	0	0	0	0	0	0	2010	2	26	1	4	8	0	0	0	0	0	0	0	0.866	0.5	0.823	0.5681	-0.4339	-0.901	1.1341	1.3266	-0.6887	0.7638	0.7685	2.1395	-3.0901	TRUE	FALSE	FALSE			
1	#####	21628	FALSE	0.2392	-0.7367	-1.6052	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10253	0.0782	0	0	0	0	0	0	2010	3	5	1	4	9	0	0	0	0	0	0	0	0	1	6.12E-17	0.8855	0.4647	-0.4339	-0.901	0.1556	1.1404	0.4156	0.729	0.6675	2.2814	0.3391	TRUE	FALSE	FALSE		
1	#####	21043	FALSE	0.2392	-0.1247	-1.5136	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1026	0.0782	0	0	0	0	0	0	2010	3	12	1	4	10	0	0	0	0	0	0	0	0	1	6.12E-17	0.935	0.3546	-0.4339	-0.901	0.2625	0.1606	1.3512	0.4462	0.5942	1.6747	-0.1079	TRUE	FALSE	FALSE		
1	#####	22137	FALSE	0.2392	-0.2987	-1.3981	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10218	0.0782	0	0	0	0	0	0	2010	3	19	1	4	11	0	0	0	0	0	0	0	0	0	1	6.12E-17	0.9709	0.2393	-0.4339	-0.901	0.2279	0.2676	1.1543	0.2296	0.549	-0.1423	0.1537	TRUE	FALSE	FALSE	
1	#####	26229	FALSE	0.2392	-0.4683	-1.3719	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10168	0.0782	0	0	0	0	0	0	2010	3	26	1	4	12	0	0	0	0	0	0	0	0	1	6.12E-17	0.9927	0.1205	-0.4339	-0.901	0.2761	0.233	0.171	0.3062	0.5382	0.0745	0.5715	TRUE	FALSE	FALSE		
1	#####	57258	FALSE	0.2392	0.1182	-1.4002	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10117	-0.0817	0	0	0	0	0	0	2010	4	2	2	4	13	0	0	0	0	0	0	0	0	0	0.866	-0.5	1	-1.61E-16	-0.4339	-0.901	0.4566	0.2813	0.2784	0.7043	0.721	2.9993	4.324	TRUE	FALSE	FALSE	
1	#####	42961	FALSE	0.2392	0.3128	-1.289	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10067	-0.0817	0	0	0	0	0	0	2010	4	9	2	4	14	0	0	0	0	0	0	0	0	0	0.866	-0.5	0.9927	-0.1205	-0.4339	-0.901	1.8246	0.462	0.2437	0.9506	0.7036	2.7923	-1.9904	TRUE	FALSE	FALSE	
1	#####	17597	FALSE	0.2392	0.3377	-1.2061	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10033	-0.0817	0	0	0	0	0	0	2010	4	16	2	4	15	0	0	0	0	0	0	0	0	0	0.866	-0.5	0.9709	-0.2393	-0.4339	-0.901	1.1943	1.8319	0.2921	0.8996	0.5679	3.0863	-3.532	TRUE	FALSE	FALSE	
1	#####	16145	FALSE	0.2392	0.2575	-1.2345	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1002	-0.0817	0	0	0	0	0	0	2010	4	23	2	4	16	0	0	0	0	0	0	0	0	0	0.866	-0.5	0.935	-0.3546	-0.4339	-0.901	0.076	1.2007	0.4734	0.7863	0.5495	3.5615	-0.2008	TRUE	FALSE	FALSE	
1	#####	16555	FALSE	0.2392	0.3968	-1.2672	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10007	-0.0817	0	0	0	0	0	0	2010	4	30	2	4	17	0	0	0	0	0	0	0	0	0	0.866	-0.5	0.8855	-0.4647	-0.4339	-0.901	0.012	0.0808	1.8482	0.329	0.5196	2.1945	0.0585	TRUE	FALSE	FALSE	
1	#####	17414	FALSE	0.2392	0.6754	-1.1472	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	0.9995	-0.0817	0	0	0	0	0	0	2010	5	7	2	4	18	0	0	0	0	0	0	0	0	0	0	0.866	0.823	-0.5681	-0.4339	-0.901	0.0301	0.1617	1.2148	0.0419	0.4991	-0.2467	0.121	TRUE	FALSE	FALSE	
1	#####	18927	FALSE	0.2392	0.7963	-1.1058	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	0.9994	-0.0817	0	0	0	0	0	0	2010	5	14	2	4	19	0	0	0	0	0	0	0	0	0	0.5	-0.866	0.7485	-0.6631	-0.4339	-0.901	0.0679	0.0348	0.091	0.0569	0.481	-0.1408	0.2121	TRUE	FALSE	FALSE	
1	#####	14773	FALSE	0.2392	0.8863	-1.1669	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10065	-0.0817	0	0	0	0	0	0	2010	5	21	2	4	20	0	0	0	0	0	0	0	0	0	0.5	-0.866	0.6631	-0.7485	-0.4339	-0.901	0.1346	0.0728	0.0267	0.0414	0.4162	-0.0419	-0.5772	TRUE	FALSE	FALSE	
1	#####	15580	FALSE	0.2392	1.1031	-1.313	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10137	-0.0817	0	0	0	0	0	0	2010	5	28	2	4	21	0	0	0	0	0	0	0	0	0	0.5	-0.866	0.5681	-0.823	-0.4339	-0.901	-0.0485	0.1395	0.0448	0.0305	0.1804	-0.0161	0.1139	TRUE	FALSE	FALSE	
1	#####	17558	FALSE	0.2392	1.1167	-1.4308	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10208	-0.0817	0	0	0	0	0	0	2010	6	4	2	4	22	0	0	0	0	0	0	0	0	0	1	6.12E-16	-1	0.4647	-0.8855	-0.4339	-0.901	-0.0129	-0.0438	0.0829	0.0321	0.0367	-0.0121	0.2769	TRUE	FALSE	FALSE
1	#####	16638	FALSE	0.2392	1.1026	-1.5115	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1028	-0.0817	0	0	0	0	0	0	2010	6	11	2	4	23	0	0	0	0	0	0	0	0	0	1	6.12E-16	-1	0.3546	-0.935	-0.4339	-0.901	0.0743	-0.0082	0.1499	0.0064	0.0313	-0.1435	-0.1268	TRUE	FALSE	FALSE
1	#####	16216	FALSE	0.2392	1.302	-1.5791	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10279	-0.0817	0	0	0	0	0	0	2010	6	18	2	4	24	0	0	0	0	0	0	0	0	0	1	6.12E-16	-1	0.2393	-0.9709	-0.4339	-0.901	0.0337	0.0791	-0.0341	0.0226	0.0317	-0.2196	-0.0573	TRUE	FALSE	FALSE
1	#####	16329	FALSE	0.2392	1.3145	-1.5442	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1025	-0.0817	0	0	0	0	0	0	2010	6	25	2	4	25	0	0	0	0	0	0	0	0	0	1	6.12E-16	-1	0.1205	-0.9927	-0.4339	-0.901	0.0151	0.0385	0.0016	0.031	0.0304	-0.263	0.0171	TRUE	FALSE	FALSE
1	#####	16333	FALSE	0.2392	1.1286	-1.5093	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	1022	-0.093	0	0	0	0	0	0	2010	7	2	3	4	26	0	0	0	0	0	0	0	0	0	0	-0.5	-0.866	#####	-1	-0.4339	-0.901	0.0201	0.0199	0.0892	0.0173	0.0243	-0.3471	0.002	TRUE	FALSE	FALSE
1	#####	17689	FALSE	0.2392	1.1053	-1.5682	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10191	-0.093	0	0	0	0	0	0	2010	7	9	3	4	27	0	0	0	0	0	0	0	0	0	0	-0.5	-0.866	-0.1205	-0.9927	-0.4339	-0.901	0.0203	0.0248	0.0485	0.0291	0.0173	-0.245	0.1902	TRUE	FALSE	FALSE
1	#####	17151	FALSE	0.2392	1.25	-1.6096	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10189	-0.093	0	0	0	0	0	0	2010	7	16	3	4	28	0	0	0	0	0	0	0	0	0	0	-0.5	-0.866	-0.2393	-0.9709	-0.4339	-0.901	0.08	0.025	0.0238	0.0396	0.0308	-0.2517	-0.0735	TRUE	FALSE	FALSE
1	#####	15360	FALSE	0.2392	1.2614	-1.6423	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10223	-0.093	0	0	0	0	0	0	2010	7	23	3	4	29	0	0	0	0	0	0	0	0	0	0	-0.5	-0.866	-0.3546	-0.935	-0.4339	-0.901	0.0563	0.0849	0.0348	0.0287	0.0295	-0.1831	-0.248	TRUE	FALSE	FALSE
1	#####	15382	FALSE	0.2392	1.179	-1.5725	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10258	-0.093	0	0	0	0	0	0	2010	7	30	3	4	30	0	0	0	0	0	0	0	0	0	0	-0.5	-0.866	-0.4647	-0.8855	-0.4339	-0.901	-0.0226	0.0611	0.035	0.018	0.0172	-0.1462	0.0044	TRUE	FALSE	FALSE
1	#####	17508	FALSE	0.2392	1.4674	-1.6009	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10292	-0.093	0	0	0	0	0	0	2010	8	6	3	4	31	0	0	0	0	0	0	0	0	0	0	-0.866	-0.5	-0.5681	-0.823	-0.4339	-0.901	-0.0217	-0.0179	0.095	0.016	0.0221	-0.1585	0.2977	TRUE	FALSE	FALSE
1	#####	15536	FALSE	0.2392	1.4587	-1.4591	-0.4279	-0.1731	-0.0847	-0.2781	-0.3952	10326	-0.093	0	0	0	0																																		

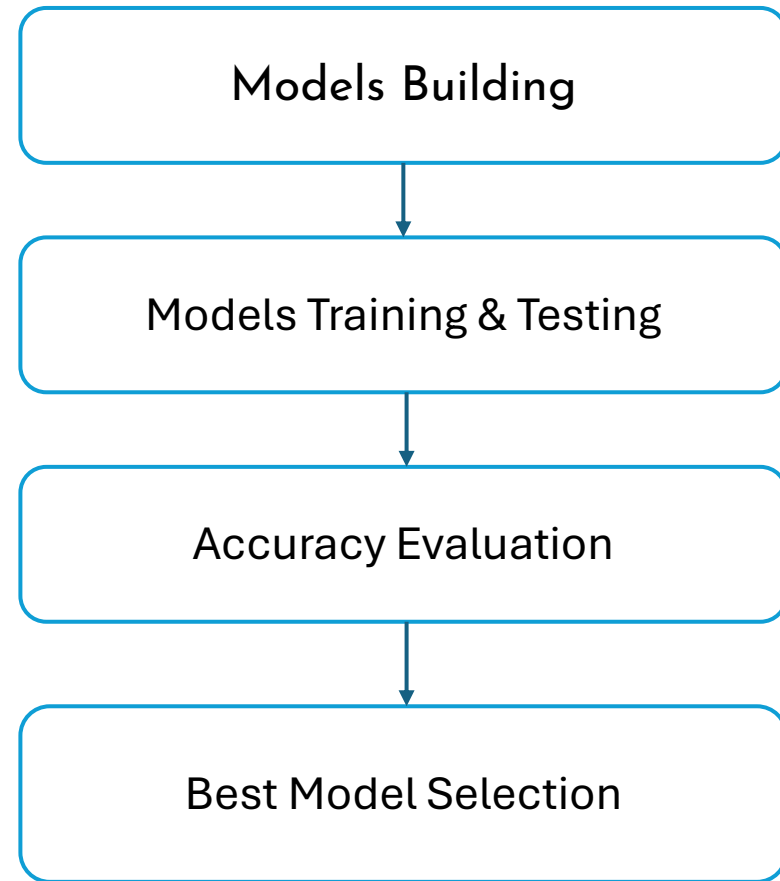
train_enhanced.csv After

[illegible]

Milestone 3: Machine learning model building

- In this Milestone, we have chosen 3 types of models and separated the features into 5 stages. Each stage contains a related group of features to make a comparison and choose the best model and the best stage of features.
- Models used → Random-Forest, XGBoost, lightGBM
- Stages used → Stage 1 (Critical) → Stage 2 (+Promotions) → Stage 3 (+Temporal) → Stage 4 (+External) → Stage 5 (+Advanced)

Milestone 3 Overview



Milestone Structure & Files Implemented

Milestone Structure

`sales_forecasting/`

<code> </code>	
<code> —</code>	<code>Config.py</code> # Configuration and hyperparameters
<code> —</code>	<code>Feature_Engineering.py</code> # Feature selection and management
<code> —</code>	<code>Evaluation.py</code> # Performance metrics and validation
<code> —</code>	<code>Models.py</code> # Model training functions
<code> —</code>	<code>Forecaster.py</code> # Main pipeline orchestration
<code> —</code>	<code>Best_model.py</code> # Best model extraction and training
<code> </code>	<code>main.py</code> # Usage examples and workflow

Progressive Feature Strategy

- Stages:
- 1) Critical → Historical patterns, Time features, Store type, Holiday
- 2) With Promotions → capture weekly and daily patterns
- 3) With Temporal → Test if promotional information improves predictions
- 4) With External → Test if external factors improve predictions
- 5) Full → Advanced features capturing sales dynamics

Progressive Feature Strategy

- Previous models were trained through 5 stages with increasing complexity



Stage	Features	Focus
1. Critical	13	Lags, rolling stats, time, store type
2. Promotions	23	Added markdown indicators
3. Temporal	37	Extended time patterns
4. External	42	Economic factors
5. Full	44	All features + momentum

```
41 # Feature groups for progressive modeling
42 FEATURE_GROUPS = {
43     'critical': [
44         'Sales_Lag1', 'Sales_Lag2', 'Sales_Lag4',
45         'Sales_Rolling_Mean_4', 'Sales_Rolling_Mean_8',
46         'Month', 'Quarter',
47         'Month_Sin', 'Month_Cos',
48         'IsHoliday',
49         'Type_A', 'Type_B', 'Type_C'
50     ],
51     'promotion': [
52         'Has_MarkDown1', 'Has_MarkDown2', 'Has_MarkDown3',
53         'Has_MarkDown4', 'Has_MarkDown5',
54         'MarkDown1', 'MarkDown2', 'MarkDown3', 'MarkDown4', 'MarkDown5'
55     ],
56     'temporal_extended': [
57         'Day', 'DayOfWeek', 'WeekOfYear',
58         'Week_Sin', 'Week_Cos', 'DayOfWeek_Sin', 'DayOfWeek_Cos',
59         'Is_Weekend', 'Is_Month_Start', 'Is_Month_End',
60         'Is_Quarter_Start', 'Is_Quarter_End', 'Is_Year_Start', 'Is_Year_End'
61     ],
62     'external': [
63         'Temperature', 'Fuel_Price', 'CPI', 'Unemployment', 'Size'
64     ],
65     'advanced': [
66         'Sales_Rolling_Std_4', 'Sales_Momentum'
67     ]
68 }
69
70 # Progressive modeling stages
71 MODELING_STAGES = ['critical', 'with_promotion', 'with_temporal', 'with_external', 'full']
--
```

Advanced Store Type Models

- Also added more specific Train separate models for each store type (A, B, C).

Type A Dominates (over half the data)

Type B Substantial (near 40 %)

Type C Smallest (only 10 %)

Store Type	Samples	Specific Model MAE	General Model MAE
Type A (Large)	215,478	\$334.64	\$106.77
Type B (Medium)	163,495	\$244.58	\$106.77
Type C (Small)	42,597	\$164.80	\$106.77

Models Evaluation & Comparison

- Evaluation Metrics used:
- MAE → Measures average prediction error
- RMSE → penalizes large errors more heavily
- R2 score → explains how well the model fits the data

```
12  ✓ def evaluate_predictions(y_true, y_pred):  
13  
14  
15     mae = mean_absolute_error(y_true, y_pred)  
16     rmse = np.sqrt(mean_squared_error(y_true, y_pred))  
17     r2 = r2_score(y_true, y_pred)  
18  
19  ✓     return {  
20         'MAE': mae,  
21         'RMSE': rmse,  
22         'R2': r2  
23     }  
24
```

Models Evaluation & Comparison

- It is obvious in the figure that all the models show massive improvement with the last feature stage (Full) stage.

According to this comparison, we chose the Random-Forest Model in the stage (Full) as the best model ready for deployment.

Stage	RF MAE	XGB MAE	LGB MAE	Ensemble MAE	Best R ²
Critical	\$952.23	\$1,019.59	\$1,119.66	\$1,000.89	0.9915
Promotions	\$956.65	\$1,099.26	\$1,202.42	\$1,041.33	0.9915
Temporal	\$936.75	\$991.32	\$1,108.92	\$974.38	0.9919
External	\$937.79	\$979.22	\$1,121.64	\$974.56	0.9919
Full	\$106.77	\$311.79	\$489.84	\$274.39	0.9996

Feature Importance Analysis

- Top 10 Features for Random-Forest Model.

- Business Implications:

- Rolling Averages (79% combined)
- Holiday Impact (4.23%)
- Promotional Limitation (<1%)

Category	Total Importance	Key Finding
Historical	86.79%	Past sales patterns dominate predictions
Temporal	8.68%	Time features supplement historical data
Advanced	2.65%	Momentum and volatility are critical
Store	1.21%	Store type has moderate impact
External	1.42%	Economic factors minimally impactful
Promotional	0.85%	Markdowns contribute little

Rank	Feature	Importance	Category	Insight
1	Sales_Rolling_Mean_4	54.32%	Historical	4-week average is strongest predictor
2	Sales_Rolling_Mean_8	24.71%	Historical	8-week average captures medium-term trends
3	Sales_Lag1	5.76%	Historical	Previous week provides context
4	IsHoliday	4.23%	Temporal	Holiday weeks show distinct patterns
5	Sales_Momentum	1.93%	Advanced	Trend acceleration matters
6	Quarter	1.24%	Temporal	Seasonal patterns
7	Is_Quarter_End	1.09%	Temporal	End-of-quarter effects
8	Day	0.94%	Temporal	Day-of-month patterns
9	Sales_Rolling_Std_4	0.72%	Advanced	Volatility indicator
10	Type_A	0.64%	Store	Store type matters

Models Pipeline

- Eventually, we made a Pipeline for running all the steps that contain:
- All_models_results() containing features importance
- Store_type_models()
- Best_model_results()

```
30 forecaster = SalesForecaster()
31
32
33 # comparing all features groups applied on all models to find the best model-----
34 # Best model after running (random_forest) in the feature stage (full) >>> all features used
35 # + feature importance to know which features drive predictions most and which features might be redundant
36 def All_models_results():
37
38     results = forecaster.progressive_modeling(df, target_col='Weekly_Sales')
39     importance = forecaster.get_feature_importance('random_forest', 'full')
40     print("\n\nTop 10 Most Important Features:")
41     print(importance.head(10))
42     return results
43
44 # Advanced models for each store type -----
45 def Store_type_models():
46
47     store_results = forecaster.train_store_type_models(df)
48     return store_results
49
50
51 # to access the best model directly-----
52 def Best_model_results():
53     results = train_best_random_forest(data, 'Weekly_Sales', None)
54     return results
55
56
57 # to run >>> All_models_results()
58 # to run >>> Store_type_models()
59 # to run >>> Best_model_results()
60 # End of ML_models/main.py -----
61
```

Model Saving

- Implemented to save the best model if the path is provided (Optional).

```
82  # Save model if path provided
83  |  if save_path:
84  |      print(f"\nSaving model to: {save_path}")
85  |      with open(save_path, 'wb') as f:
86  |          pickle.dump(model, f)
87  |      print("Model saved successfully")
```

Stage 4: MLOps, Deployment & Monitoring

Stage Overview

- Built an end-to-end ML system for sales forecasting
 - Random Forest model achieving 99.96% R^2 accuracy
 - Full MLOps pipeline: API, dashboard, monitoring, Docker, MLflow

Prediction Engine

- 121MB Random Forest model with 44 engineered features
 - Integrated historical data (50K+ records)
 - Dynamic path resolution for stable imports
 - Graceful handling of missing values and edge cases

REST API (FastAPI)

- Endpoints: health, predict, batch, multi-week, model info
 - Automatic Swagger UI documentation
 - Input validation with Pydantic
 - Production startup via uvicorn runner

Dashboard (Streamlit)

- 4 pages: Prediction, Performance, Monitoring, Model Info
 - Interactive visualizations with Plotly
 - Forecasting up to 52 weeks
 - Business-friendly interface

MLOps & Tracking

- MLflow tracking with 50+ experiments
 - Versioned models & registry
 - Automated experiment runner
 - Comprehensive training logs and metrics

Monitoring System

- Latency & error monitoring
 - Drift detection (KS test, distribution tracking)
 - Automated alerts and logs

Docker Deployment

- Multi-service architecture: API, Dashboard, MLflow
 - Optimized Dockerfile with caching & retries
 - docker-compose for orchestration
 - Production setup with Nginx, SSL, load balancing

Git LFS & Model Storage

- 121MB model stored via Git LFS
 - Efficient cloning and deployment
 - Version control for large binary artifacts

Cloud Deployment

- Streamlit Cloud for free deployment
 - GitHub auto-deployments
 - Fixed dependency conflicts and watcher issues
 - Live URL with HTTPS

Testing & Validation

- Unit, integration, and end-to-end tests
 - Automated verification scripts
 - Feature engineering validation

Final Architecture

- API + Dashboard + Predictor + Model + MLOps stack
 - Real-time predictions with monitoring
 - Separation of services for scalability

Achievements

- 2000+ lines of production code
 - 5500+ lines of documentation
 - Complete production-ready ML system
 - Successfully deployed live dashboard